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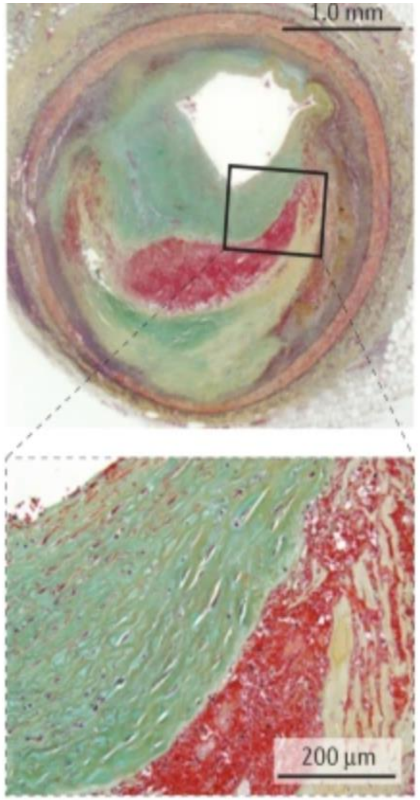
Accelerating Chest Pain:

Acute Coronary Syndromes



Hung-Fat Tse, MD, PhD

Chair, Professor of Cardiovascular Medicine
William MW Mong Professor in Cardiology
Chairperson and Chief, Department of Medicine
Chief, Cardiology Division
Queen Mary Hospital/SZ-HKU/GHK
School of Clinical Medicine, LKS Faculty of Medicine
The University of Hong Kong, Hong Kong

2023

CTM, M/51 (1)

C/O: Presents for sudden onset of chest pain at rest for 1 day

- Hx of stable CAD and hyperlipidemia on aspirin/statin and betaloc
- Defaulted medications for 6 months as no chest pain
- Constricting in character and cannot relieve by TNG

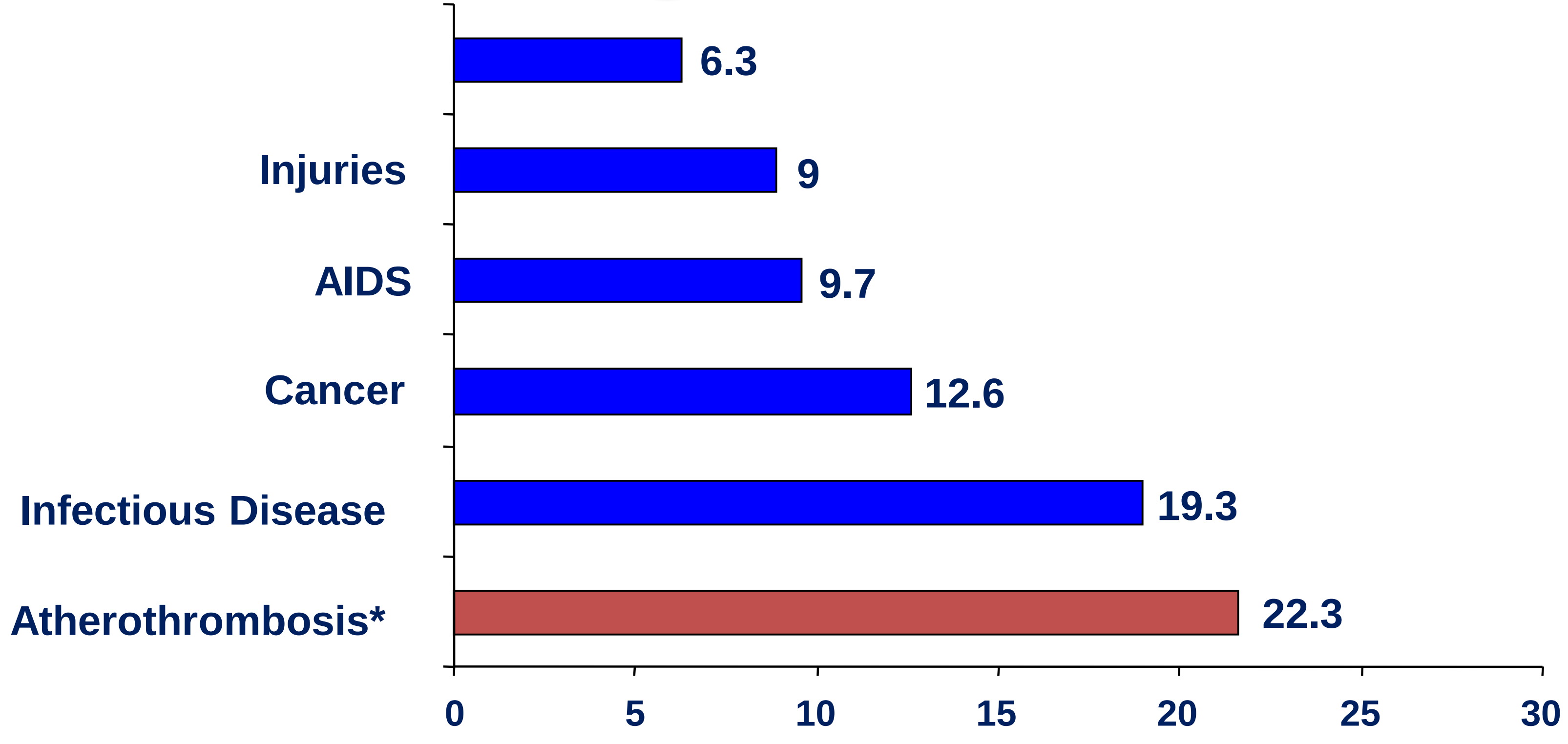
Family Hx: Father died of CAD at age of 65

Social Hx: Chronic smoker 1 pack per day and non-drinker

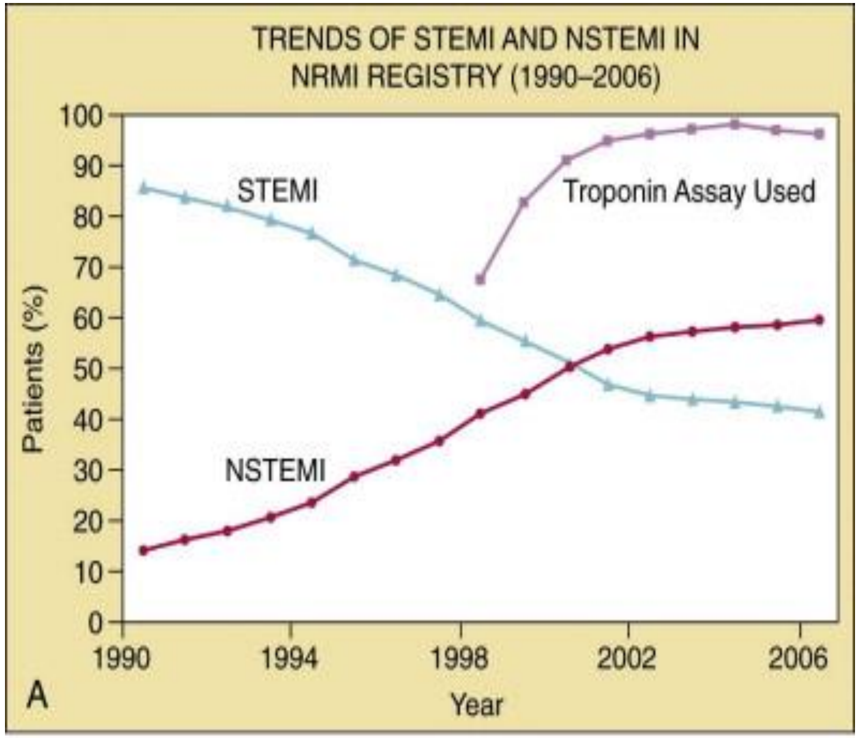
Non-ST-Segment Elevation Acute Coronary Syndrome

- **Epidemiology**
- Pathophysiology
- Diagnosis and Assessment
- Initial Therapies and Management
- Long-term Management

Atherothrombosis* is the Leading Cause of Death Worldwide[1]



Causes of Mortality (%)



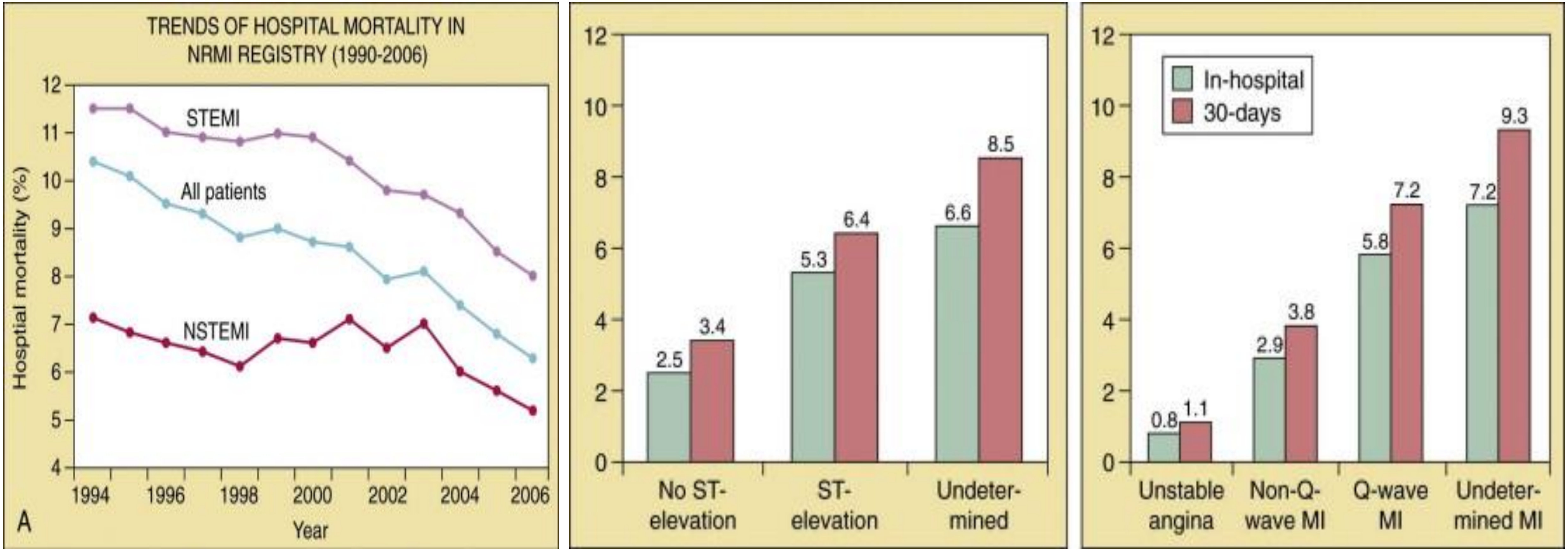
Acute Coronary Syndromes: A Companion to Braunwald's Heart Disease , 2rd Edition

*Atherothrombosis defined as ischemic heart disease and cerebrovascular disease.

[1] *The World Health Report 2001*. Geneva: WHO; 2001.

Reprod. with permission from Cannon CP. Atherothrombosis slide compendium. Available at: www.theheart.org.

Epidemiology of ACS



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Non-ST-Segment Elevation Acute Coronary Syndrome

- Epidemiology
- **Pathophysiology**
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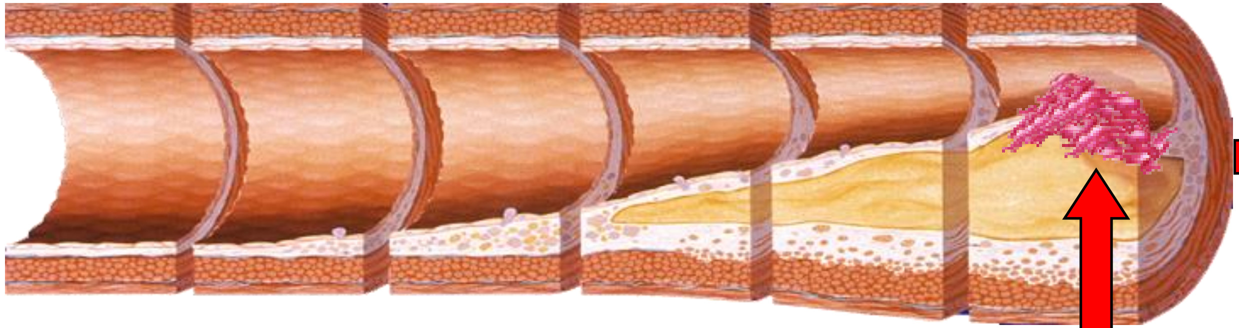
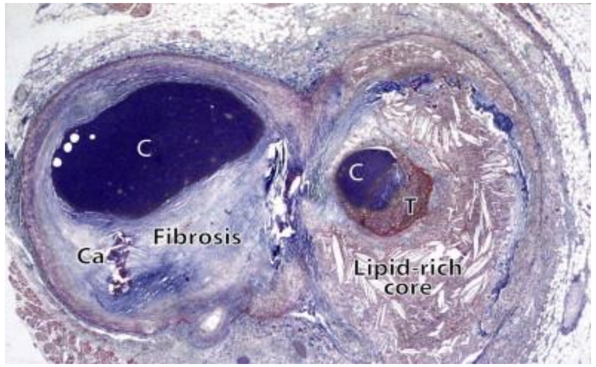
Atherothrombosis:
A Generalized and Progressive Process

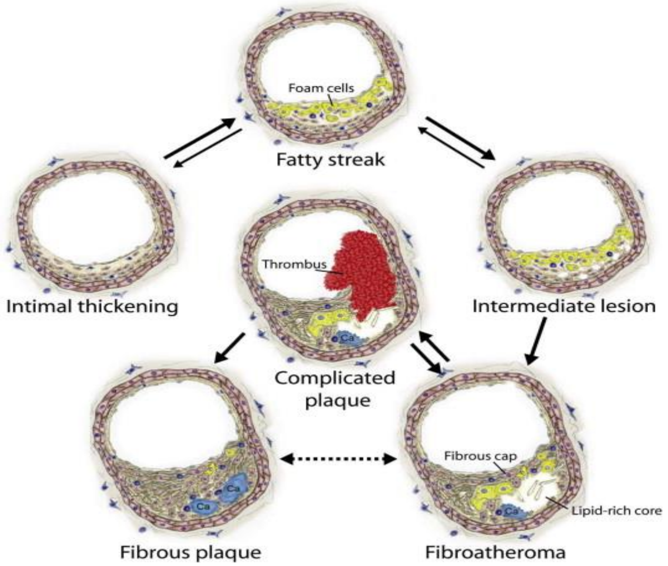


Unstable angina
MI
} **ACS**
Ischemic stroke/TIA
Critical leg ischemia

Atherosclerosis → **Atherothrombosis**

Stable angina
Intermittent claudication





Adapted from Stary HC et al. *Circulation*. 1995; 92: 1355–74, and Fuster V *et al*. *Vasc Med*. 1998; 3: 231–9.
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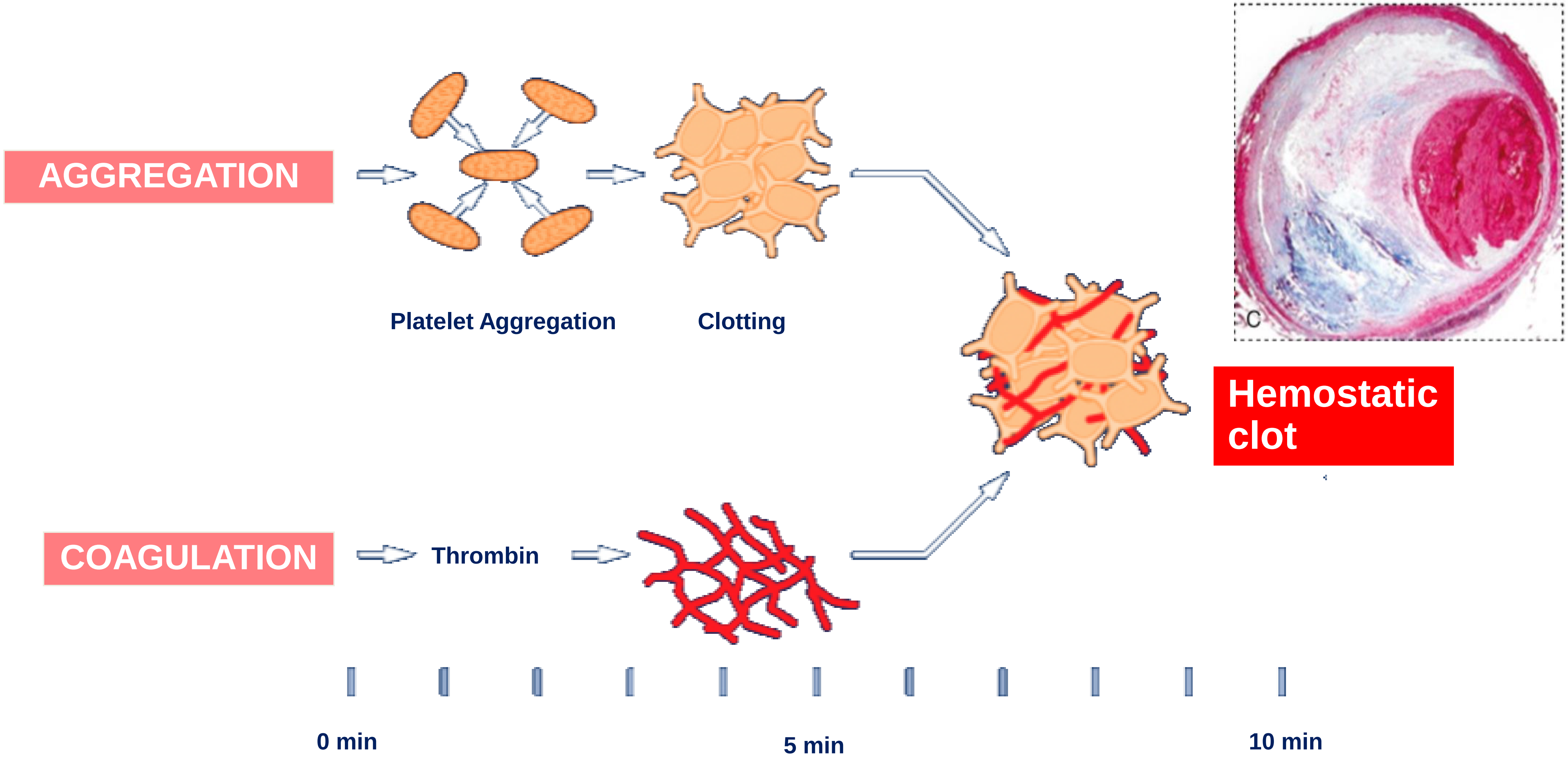
Pathophysiology of Atherothrombosis



Non-ST-Segment Elevation Acute Coronary Syndrome

- Epidemiology
- Pathophysiology
- **Diagnosis and Assessment**
- Initial Therapies and Management
- Long-term Management

Events in ACS

Atherothrombosis

↓

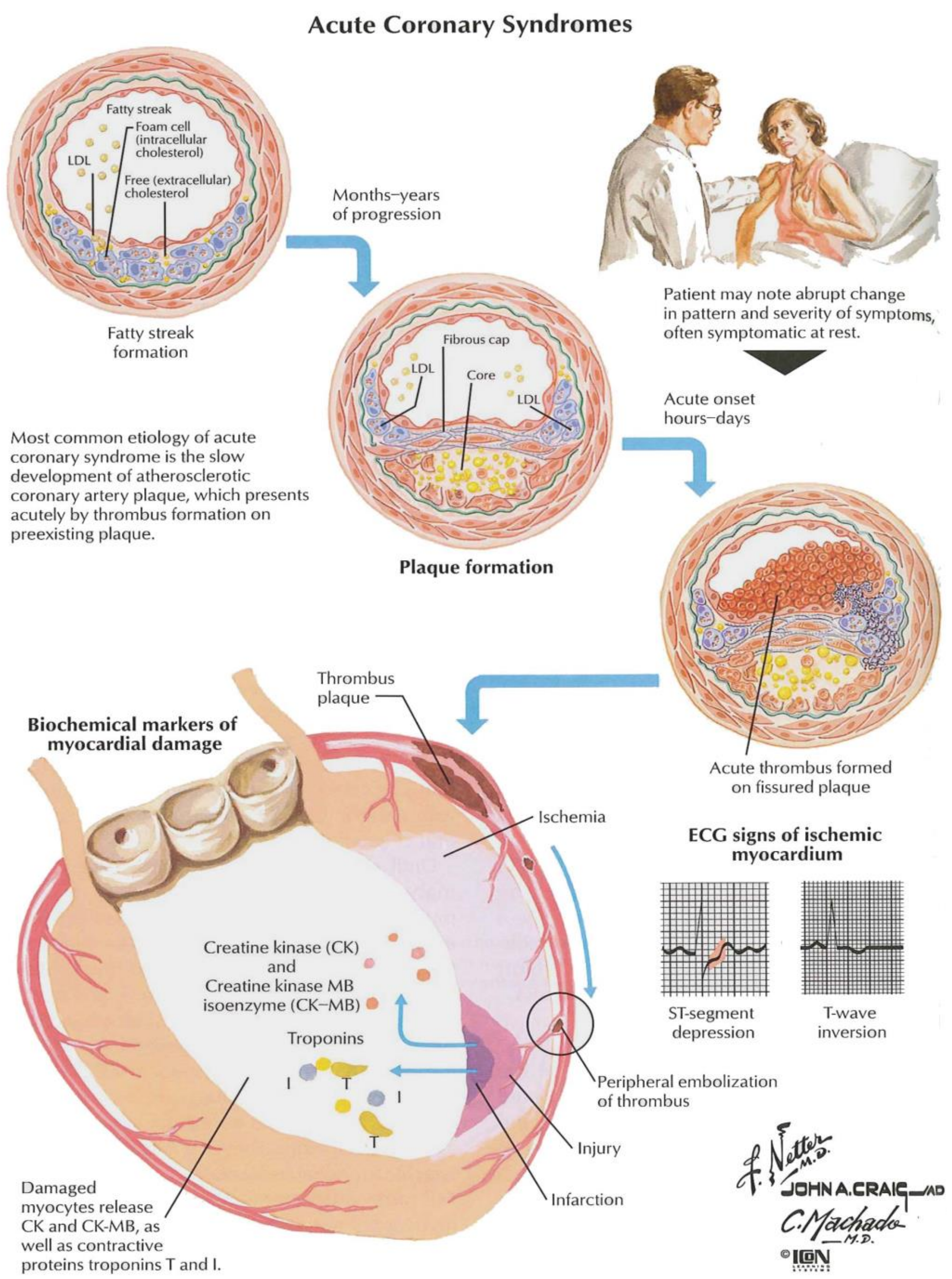
Chest pain/SOB/Collapse

↓

ECG Changes

↓

+/- Cardiac enzyme change



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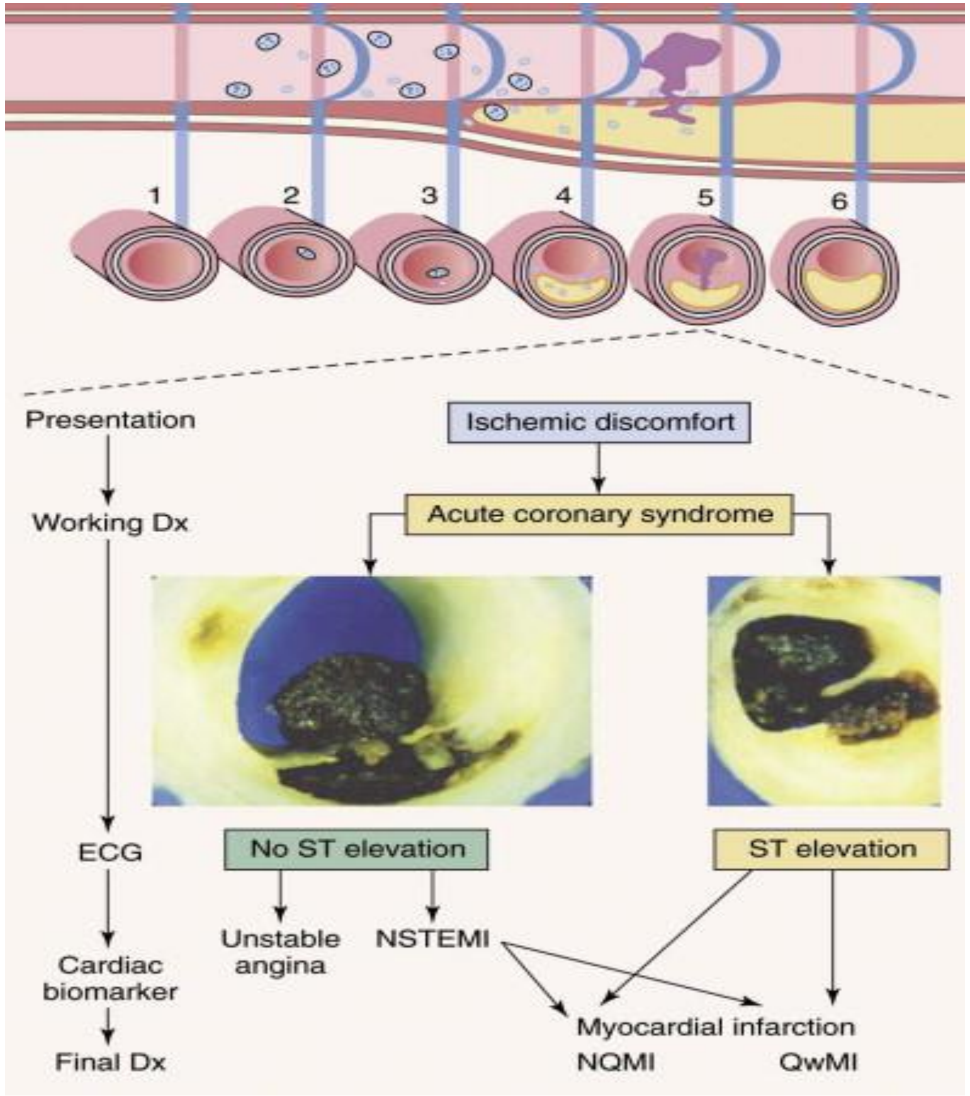
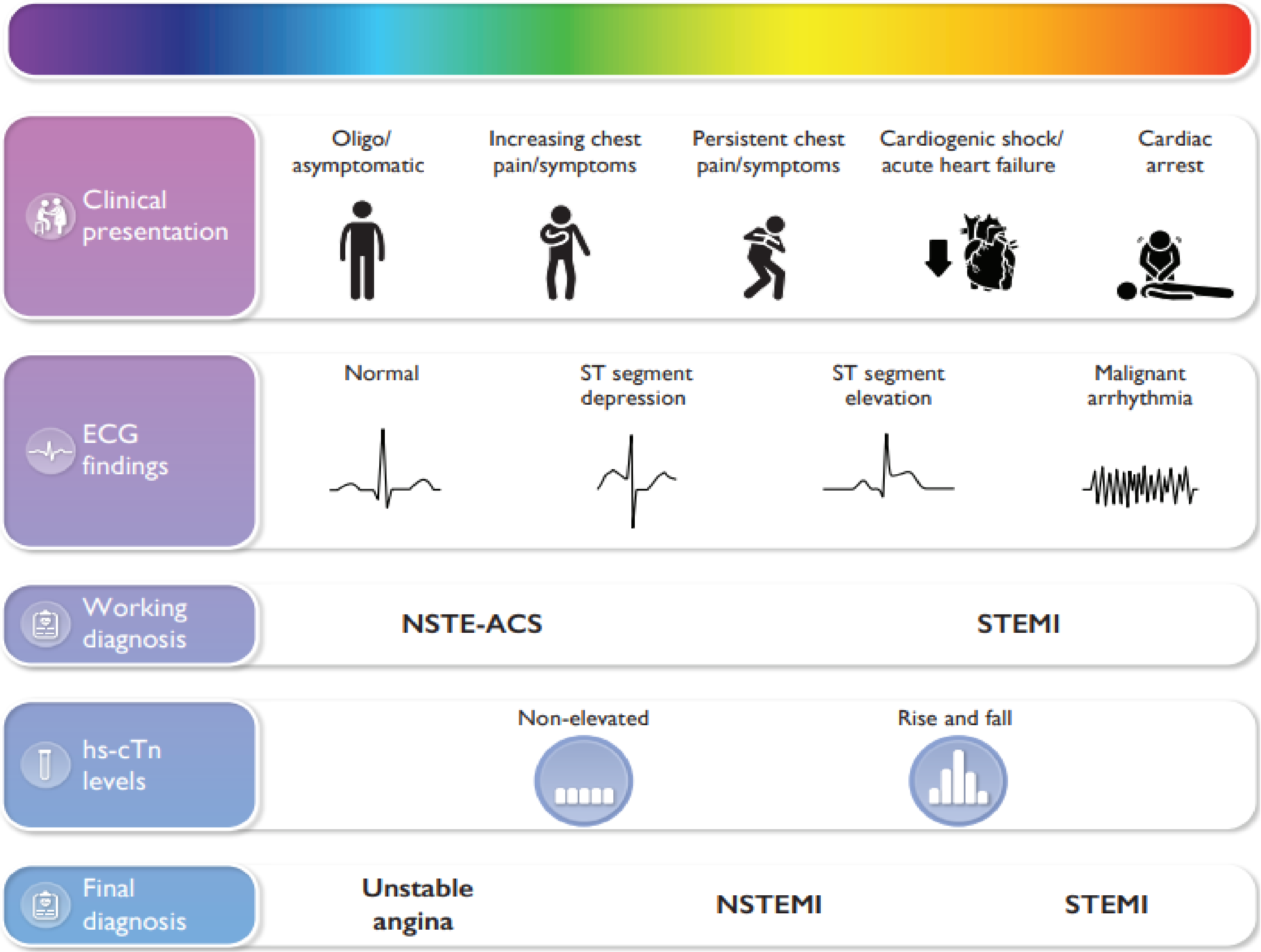
Clinical Spectrum of Acute Coronary Syndrome





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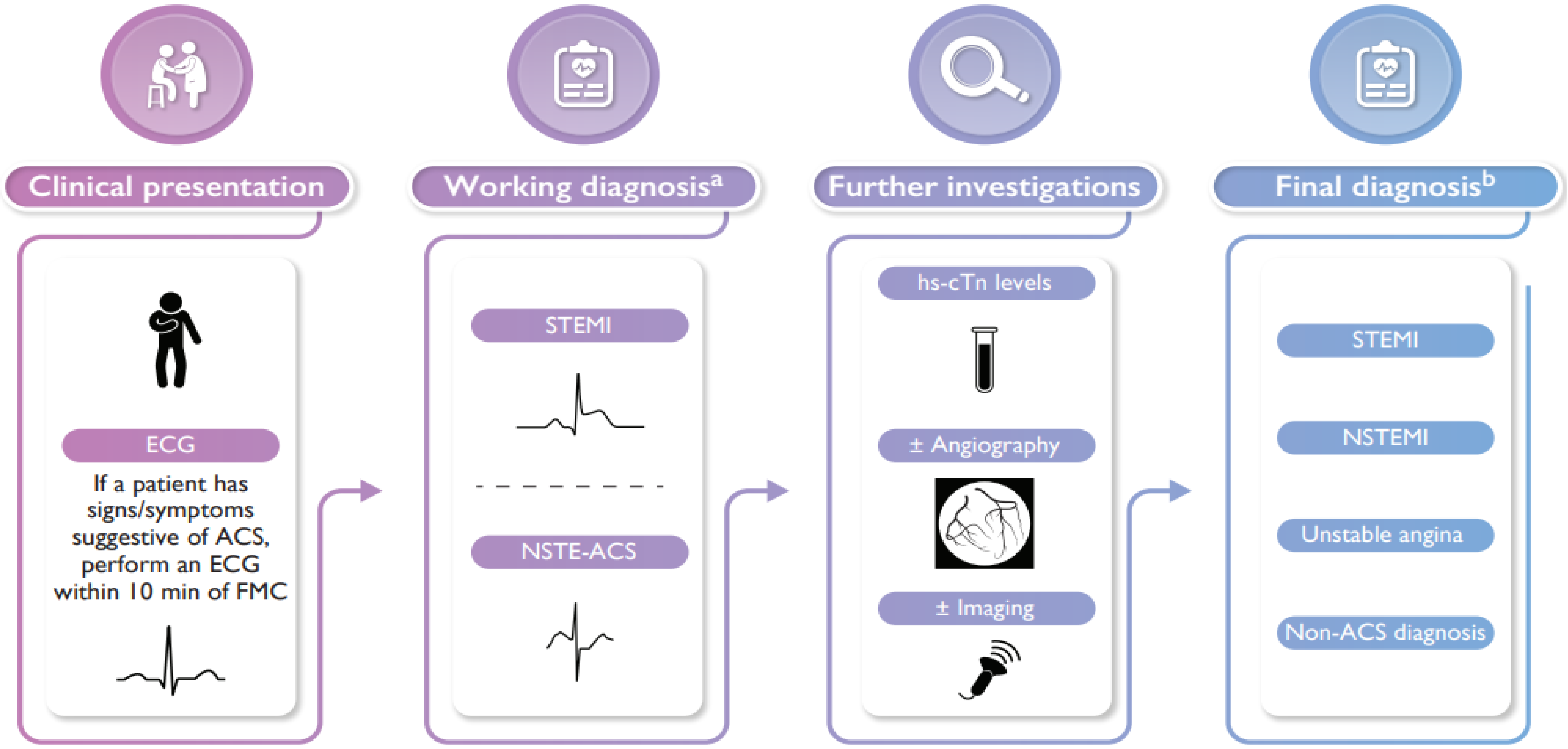
Diagnosis of Acute Coronary Syndrome



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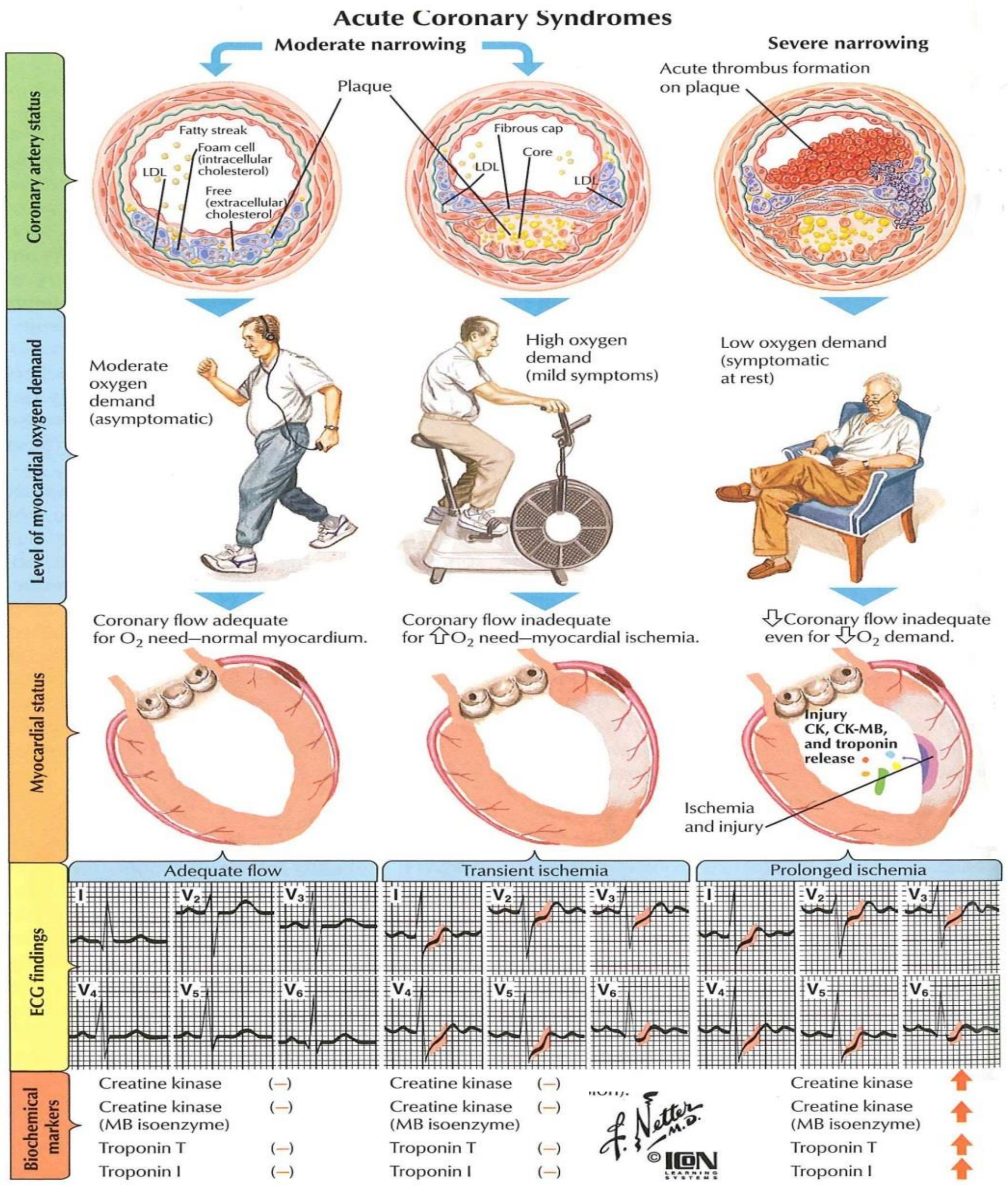
NSTE-ACS

Progression of Symptoms

- ➢ Non-occlusive thrombus on pre-existing plaque
- ➢ Dynamic obstruction (coronary spasm or vasoconstriction)
- ➢ Progressive mechanical obstruction
- ➢ Increase demand
- ➢ ?Infection/ Inflammation



NSTE-ACS

(UA/ Non-Q wave MI)

Principle Presentation

Rest Angina*	Angina occurring at rest and prolonged, usually > 20 minutes
New-onset Angina	New-onset angina of at least CCS Class III severity
Increasing Angina	Previously diagnosed angina that has become distinctly more frequent, longer in duration, or lower in threshold (i.e., increased by $\geq$ 1 CCS) class to at least CCS Class III severity.
Post-MI Angina	Recurrent of angina after a recent MI

* Patients with NSTEMI usually present with angina at rest

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Braunwald E. Circulation 1989

ACS: Principle Presentation

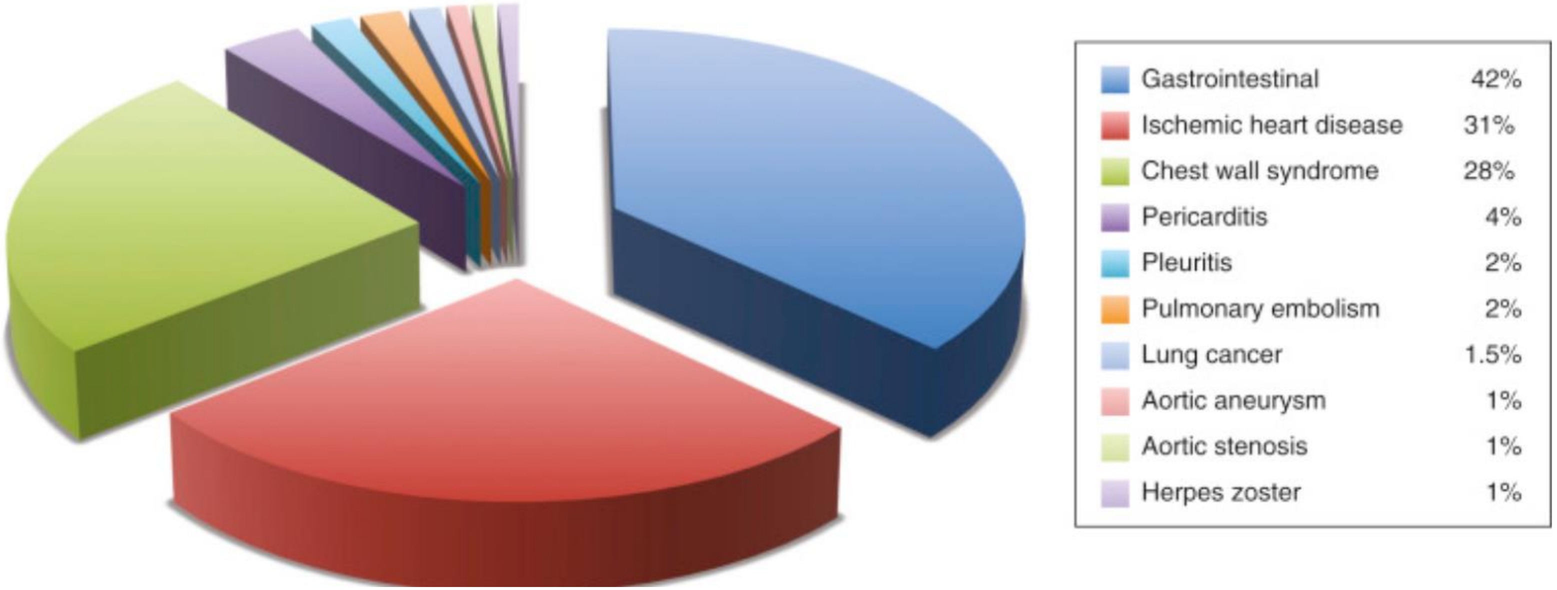
Table 1
The spectrum of acute coronary syndromes.

Unstable angina	Non-ST-Segment Elevation Myocardial Infarction (NSTEMI)	ST-Segment Elevation Myocardial Infarction (STEMI)
Anginal pain with at least one of the following features: • Is of new onset and severe • Occurs at rest or with minimal exertion • Pain is worsening in the severity and length of each episode (i.e. occurring in a crescendo pattern)	Characterized by clinical features of unstable angina in addition to elevated cardiac markers. Cardiac markers are elevated as a result of myocardial necrosis	Characterized by clinical features of myocardial infarction in addition to ST-segment elevation on a 12-lead EKG

Differential Diagnosis of ACS

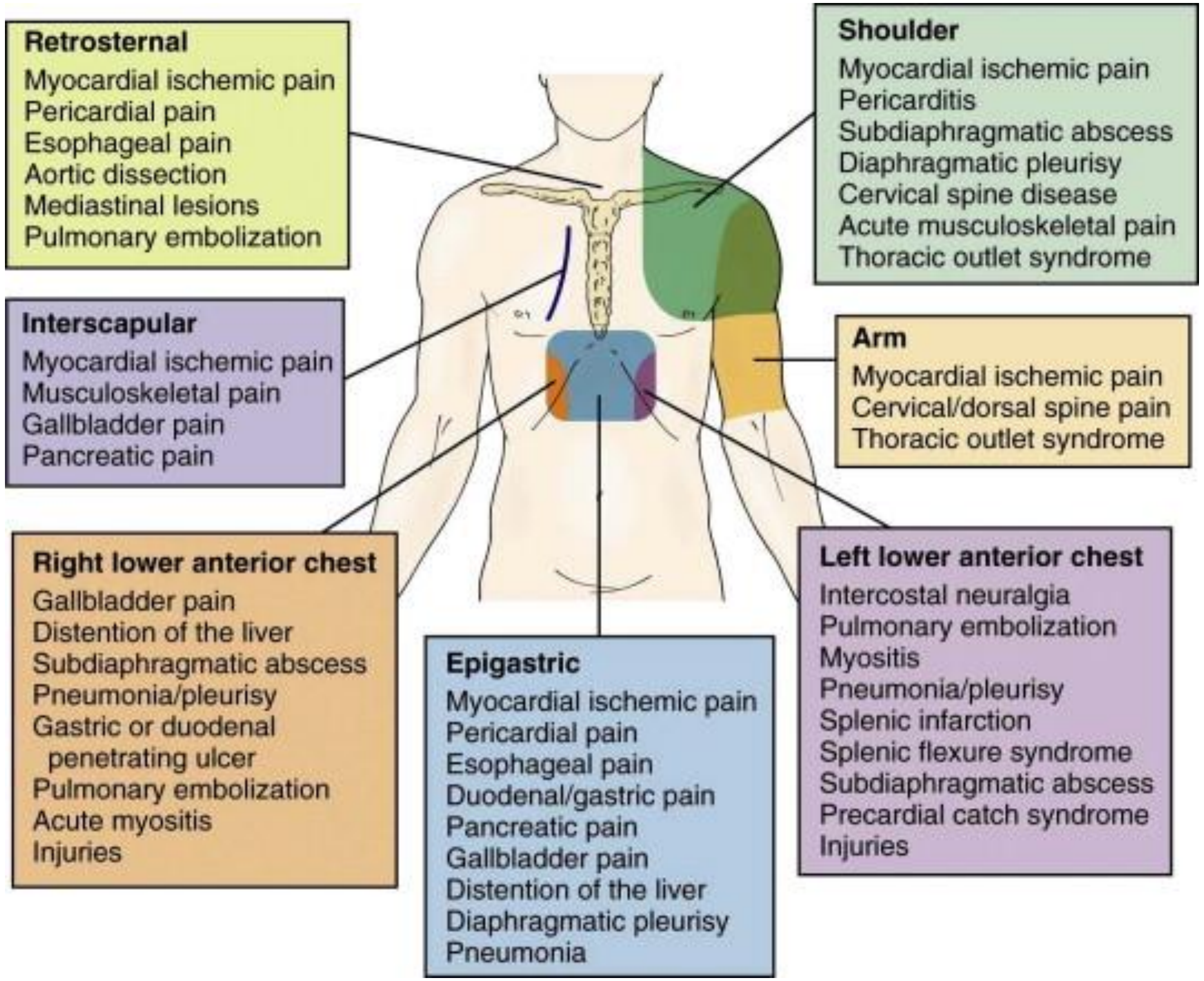
Table 6 **Differential diagnoses of acute coronary syndromes in the setting of acute chest pain**

Cardiac	Pulmonary	Vascular	Gastro-intestinal	Orthopaedic	Other
Myopericarditis	**Pulmonary embolism**	**Aortic dissection**	**Oesophagitis, reflux, or spasm**	**Musculoskeletal disorders**	**Anxiety disorders**
Cardiomyopathies[a]	**(Tension)-pneumothorax**	Symptomatic aortic aneurysm	Peptic ulcer, gastritis	Chest trauma	Herpes zoster
Tachyarrhythmias	Bronchitis, pneumonia	Stroke	Pancreatitis	Muscle injury/inflammation	Anaemia
Acute heart failure	Pleuritis		Cholecystitis	Costochondritis	
Hypertensive emergencies				Cervical spine pathologies	
Aortic valve stenosis					
Takotsubo syndrome					
Coronary spasm					
Cardiac trauma					

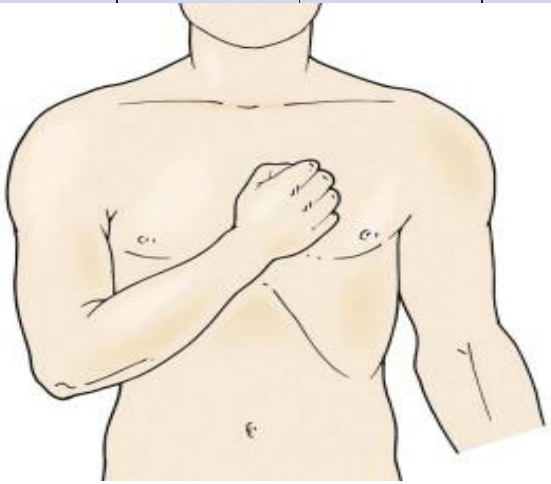


ESC 2020
Fruergaard P, et al. Eur Heart J 1996

Differential Diagnosis of ACS



Cardiac	Pulmonary	Haematological	Vascular	Gastro-intestinal	Orthopaedic/ infectious
Myocarditis	Pulmonary embolism	Sickle cell crisis	Aortic dissection	Oesophageal spasm	Cervical discopathy
Pericarditis	Pulmonary infarction	Anaemia	Aortic aneurysm	Oesophagitis	Rib fracture
Cardiomyopathy	Pneumonia Pleuritis		Cerebrovascular disease	Peptic ulcer	Muscle injury/ inflammation
Valvular disease	Pneumothorax			Pancreatitis	Costochondritis
Tako-Tsubo cardiomyopathy				Cholecystitis	Herpes zoster
Cardiac trauma					



Acute Coronary Syndromes: A Companion to Braunwald's Heart Disease , 2rd Edition

CTM, M/51 (2)

P/E: Body weight 80 kg, height 165cm (body mass index: 29),
Waist circumference: 93cm

Blood pressure 98/65mmHg, HR 102 bpm

HS-S3+ve, nil murmur

Chest-clear

Ix:

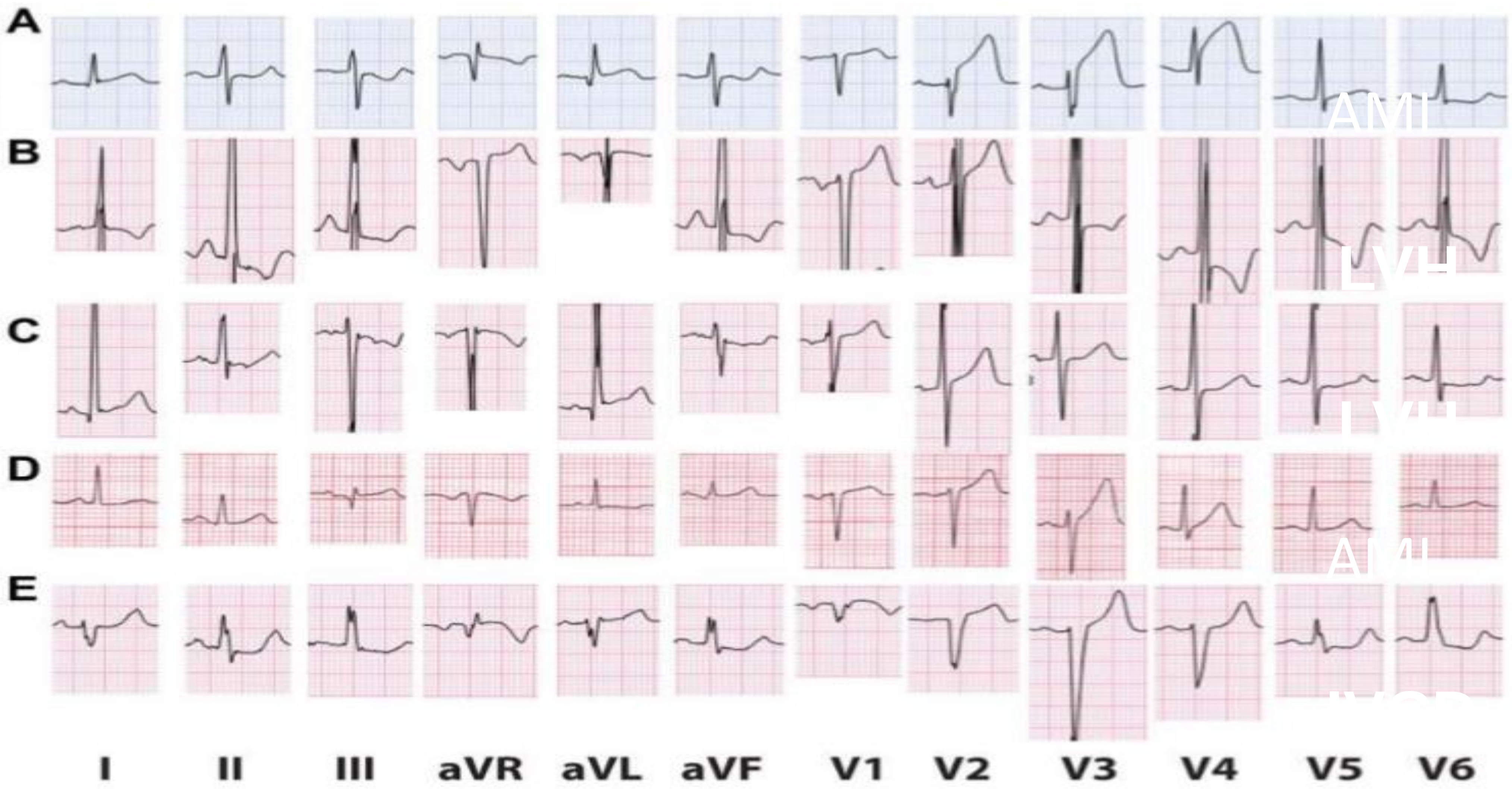
- ECG: biphasic T wave inversion over depression over V1-V6
- FBS: 6.2 mmol/l
- Lipid: TG- 2 mmol/l, HDL- 1.0 mmol/l, LDL- 4.6 mmol/l,
- hsTnT: 100

Physical Examination:

Goal	Category	Findings/Comments
Assessing risk factors	Blood pressure	>15 mm Hg arm blood pressure disparity = increased risk for peripheral arterial disease (PAD) and cardiovascular death
	Weight	Body mass index >30 kg/m^2 = obesity
	Lipid abnormalities	Cutaneous xanthomas Xanthelasma on the eyelids Corneal arcus
	Diabetes mellitus	Acanthosis nigricans Skin tags
	Tobacco abuse and chronic obstructive pulmonary disease (COPD)	Odor of tobacco Staining of teeth, fingers, or nails Premature skin wrinkling Prolonged expiration, wheezing, distant breath sounds
	Miscellaneous	Ear lobe creases (Frank sign)
Assessing for complications	Congestive heart failure	Jugular venous distention, S_3, S_4, displaced point of maximal impulse, hepatomegaly, pulmonary/peripheral edema Ischemic mitral regurgitation Low-output cardiac failure
	Arrhythmias	Ectopy, atrial fibrillation
	PAD	Carotid bruits Peripheral pulses Peripheral skin discoloration or hair loss
Assessing for other causes of angina and dyspnea	Aortic stenosis	Late peaking systolic murmur Pulsus parvus et tardus
	Hypertrophic cardiomyopathy	Harsh crescendo-decrescendo systolic murmur dynamic with provocation
	Pulmonary hypertension	Loud P2, right S4, TR murmur, RV lift/heave

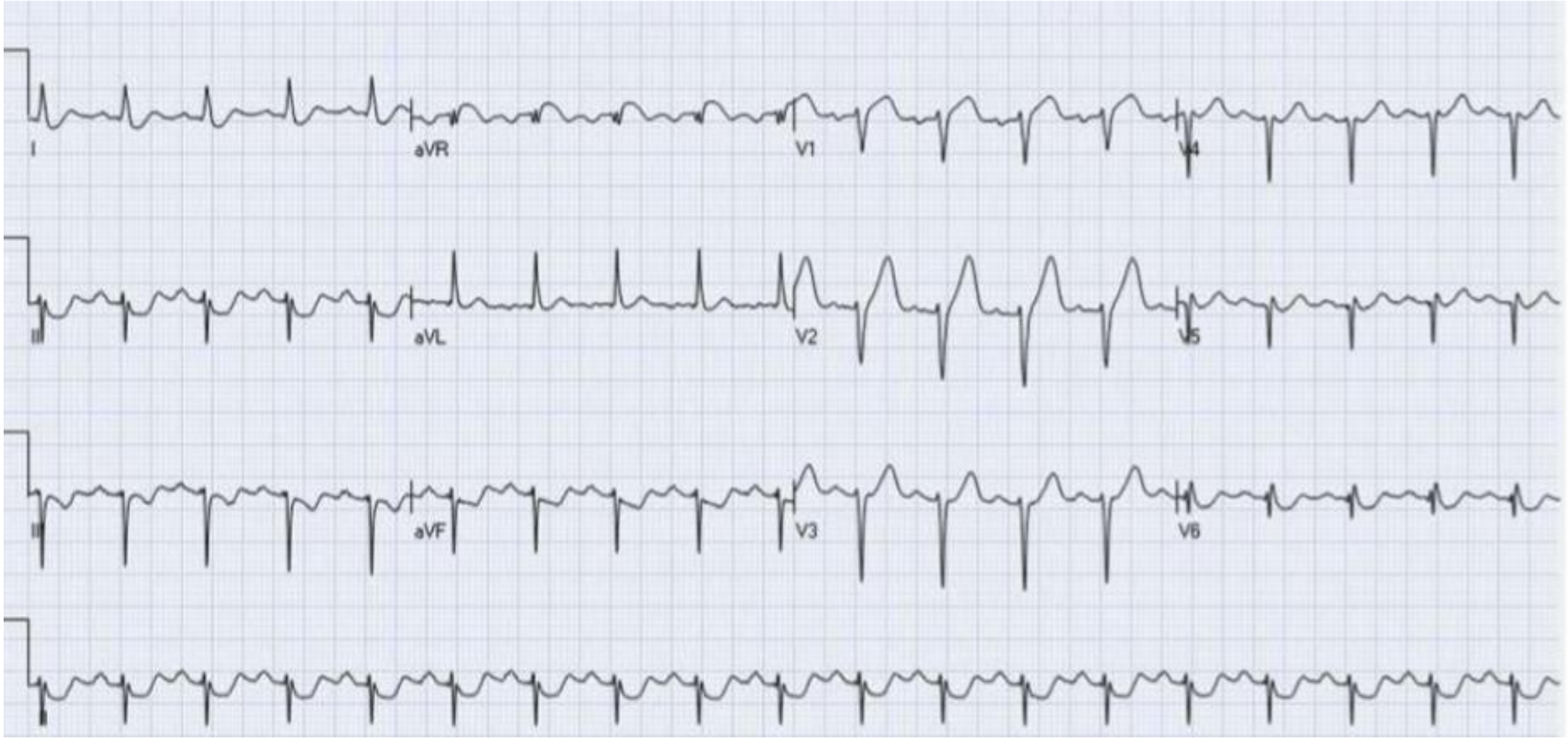
Chronic Coronary Artery Disease: A Companion to Braunwald's Heart Disease

ECG for ACS



Birubaum Y, et al. Annals of Non-invasive Electrocardiography

ECG for ACS



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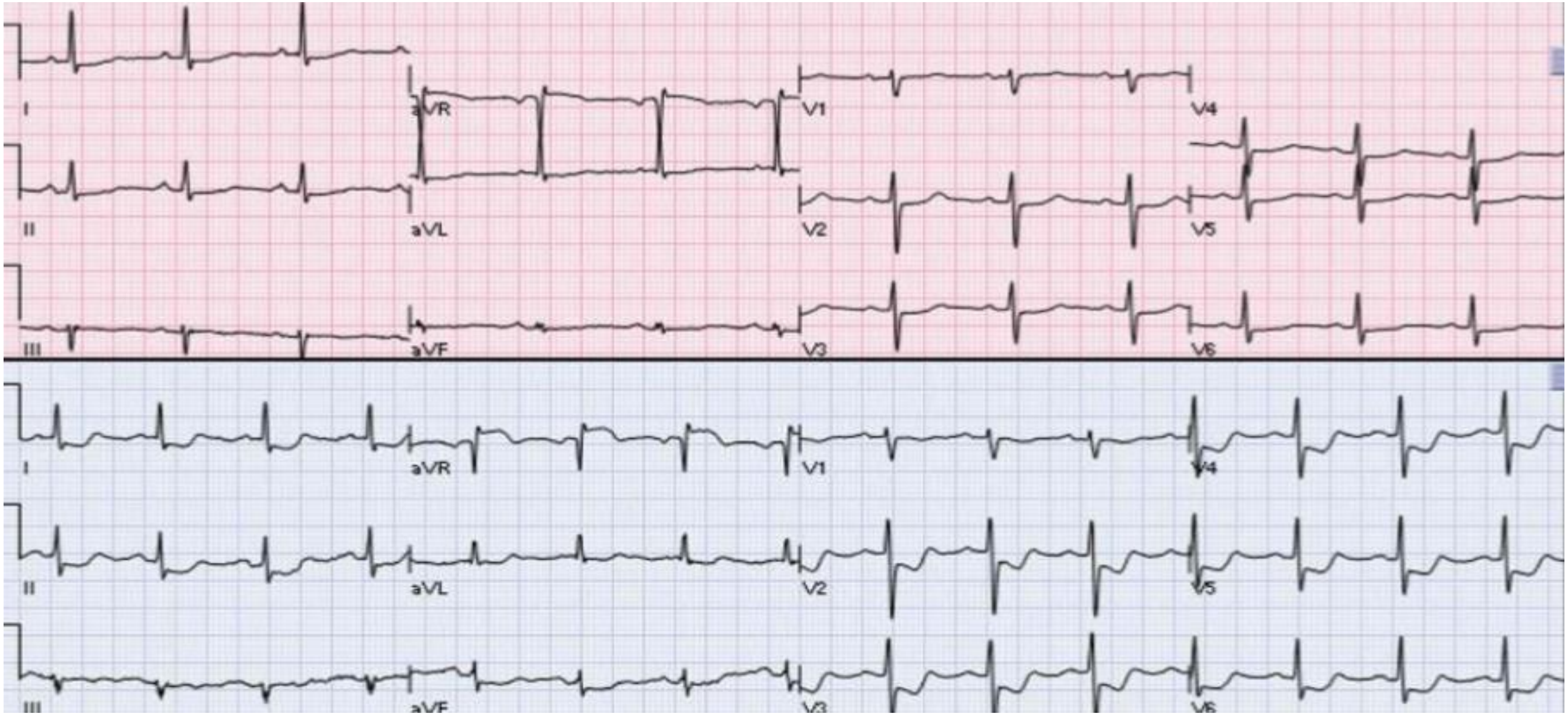
ECG for ACS



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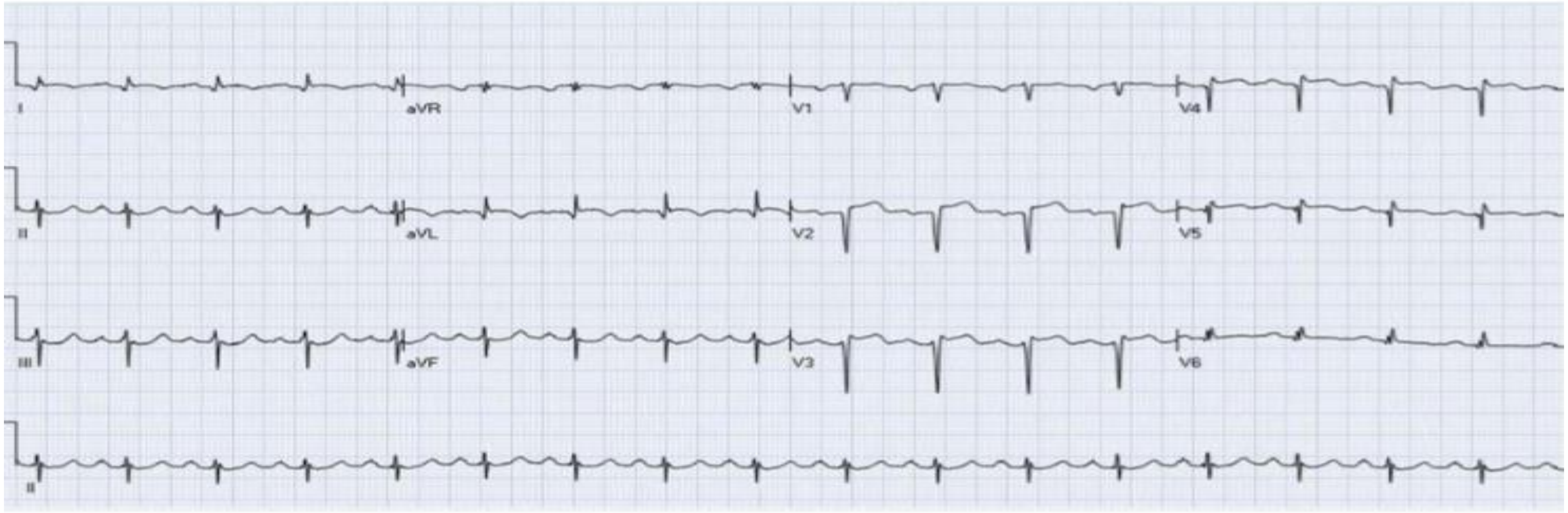
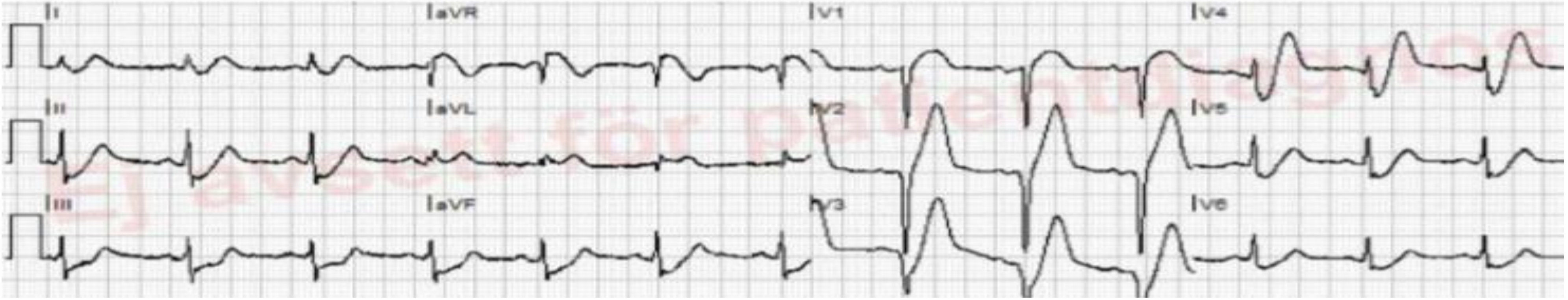
Birubaum Y, et al. Annals of Non-invasive Electrocardiography

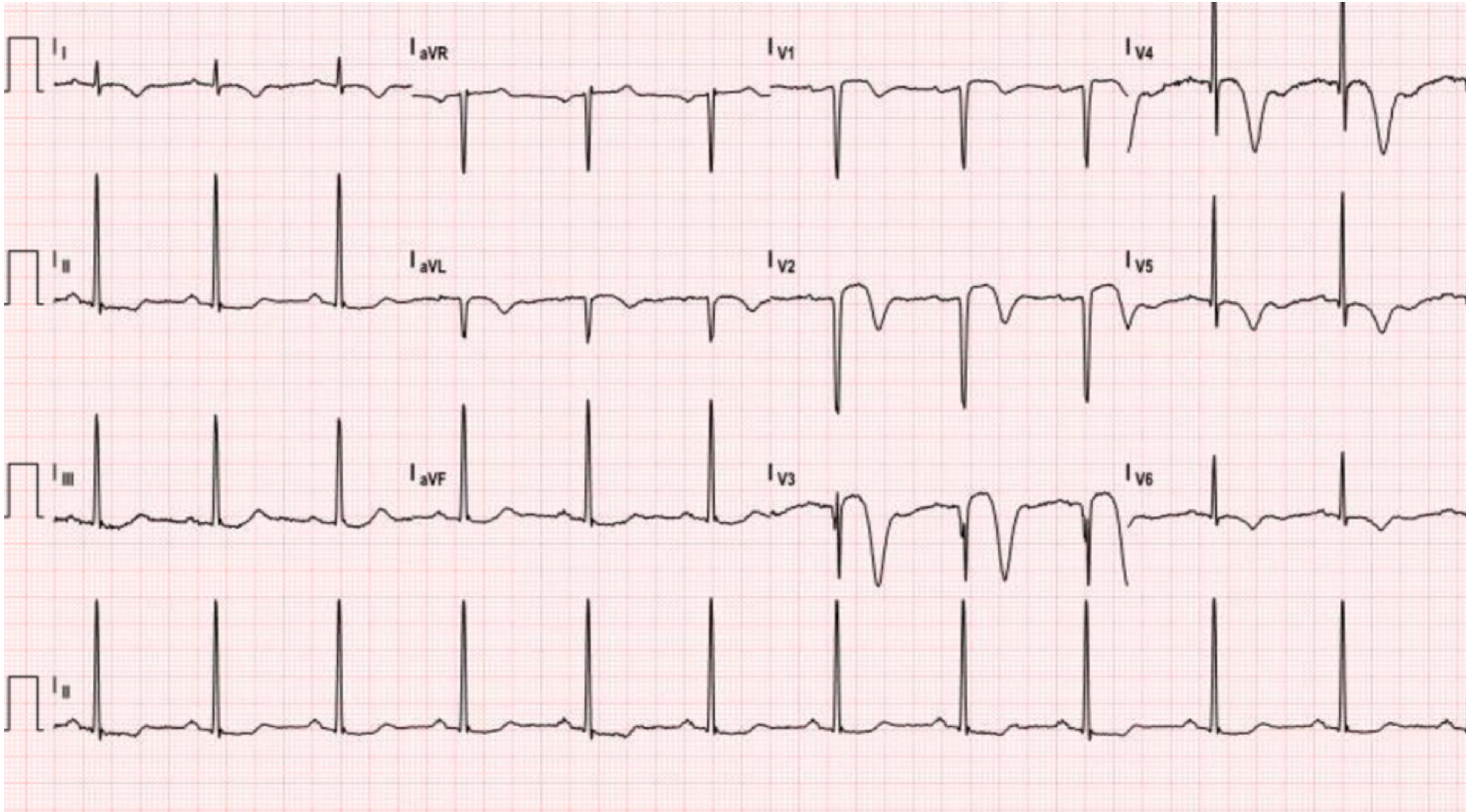
ECG for ACS



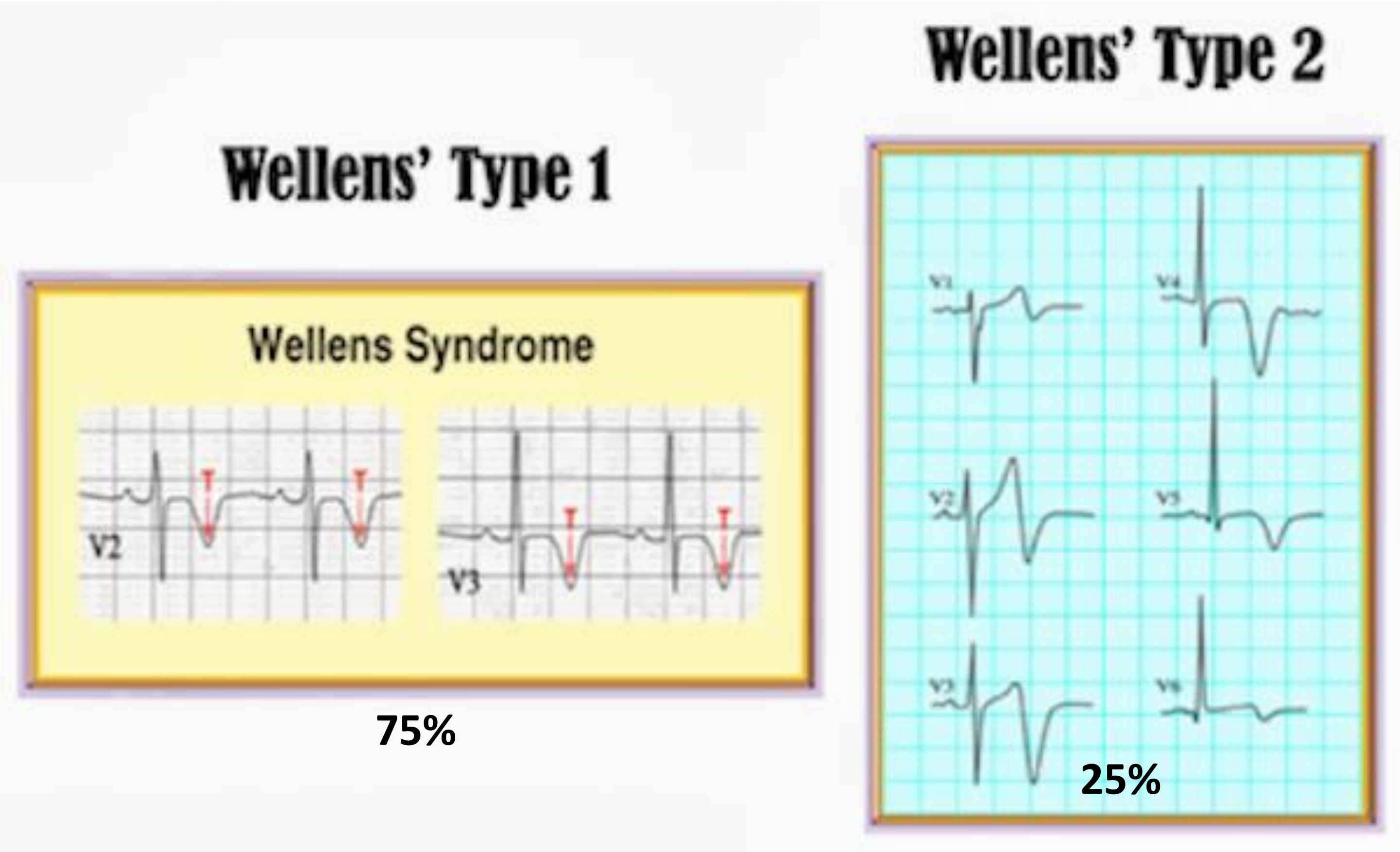


Birubaum Y, et al. Annals of Non-invasive Electrocardiography

CTM, M/51 (3)



ECG for ACS



This is a pre-infarction stage of coronary artery disease, also referred to as LAD coronary T-wave syndrome. The syndrome criteria include the following:

•History of angina + T wave inversion or biphasic t waves in V2–V4

•Normal or minimally elevated cardiac enzymes

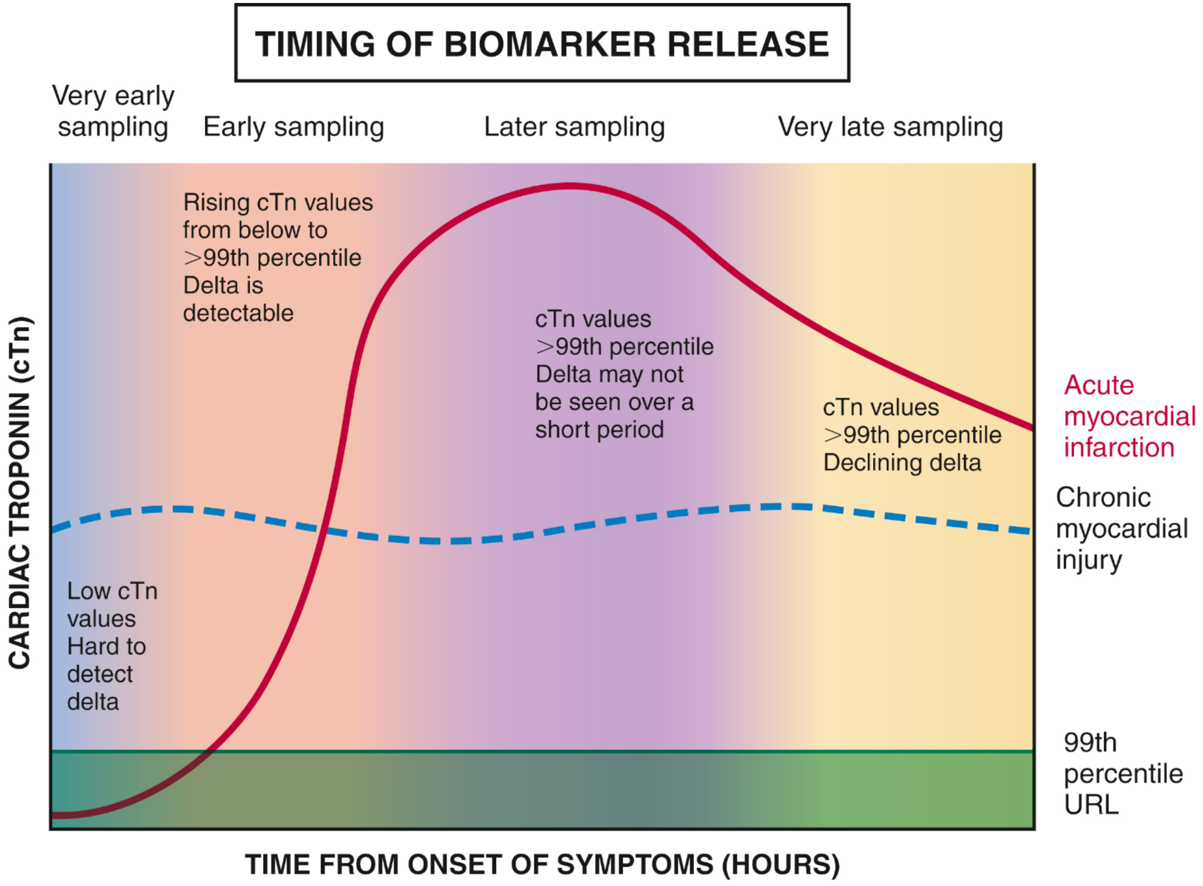
•No pathologic precordial q waves or loss of precordial R wave progression

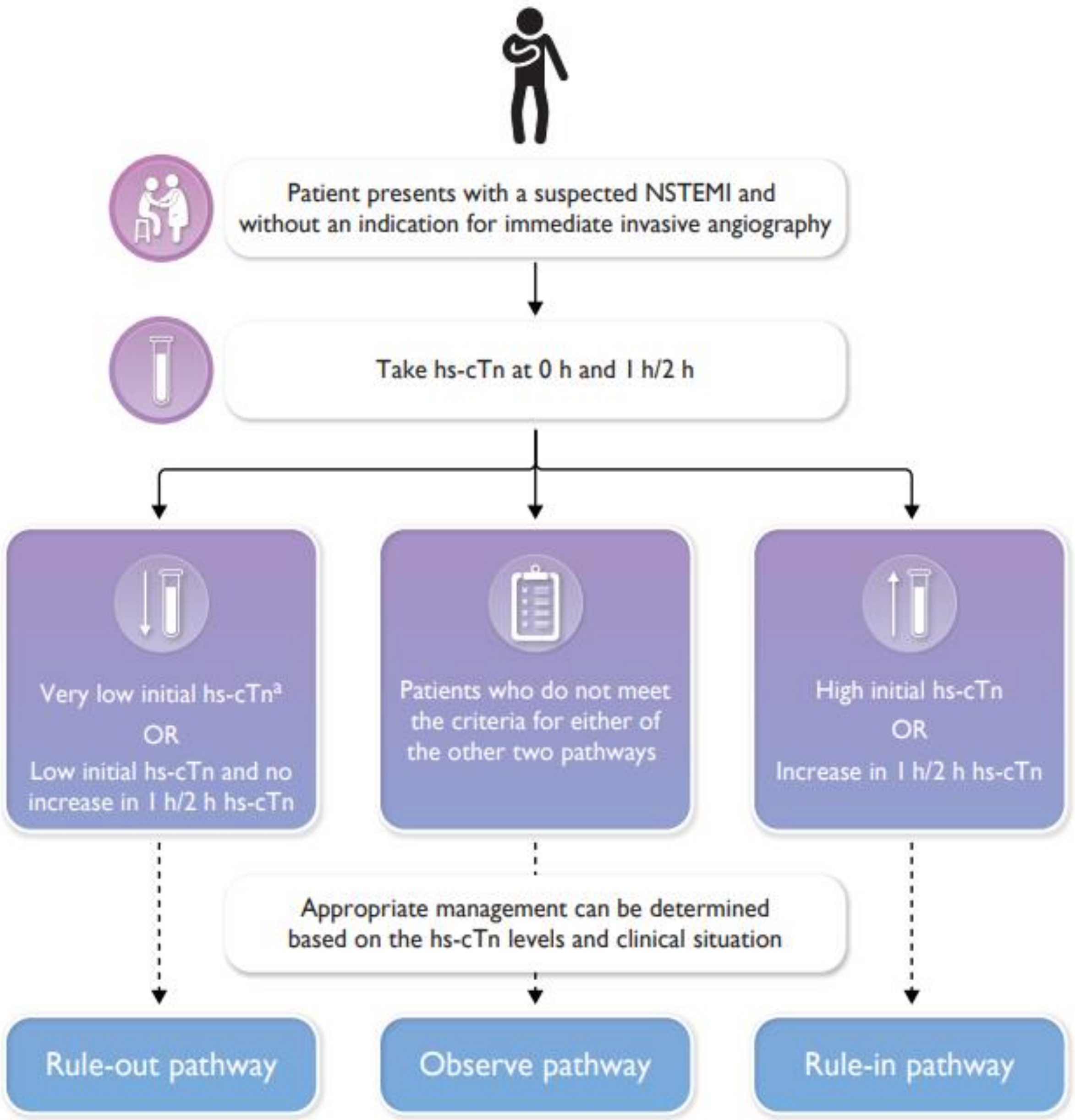
Acute MI occurs within a mean of **6 – 8.5 days after admission**

Acute MI occurs within a mean of **21.4 days after symptoms**

Wellen's Syndrome was first described in 1982 in which 75% of patients with t wave inversions in V2-V4 went on to have an acute myocardial infarction (MI). This was again repeated in 1989, and showed that all patients with this morphology had >50% LAD stenosis. The incidence in the United States is about 10-15%.

Hs-Troponin for Early Diagnosis of ACS





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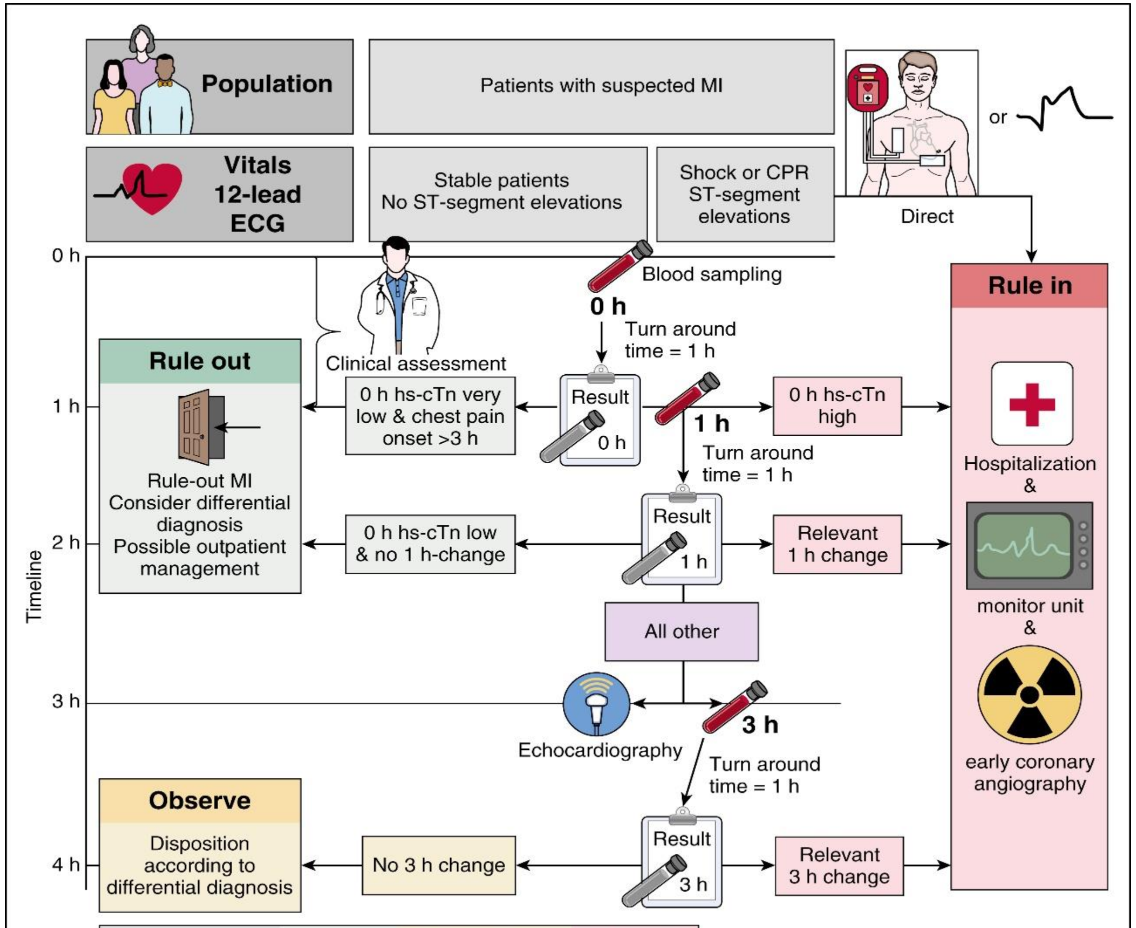
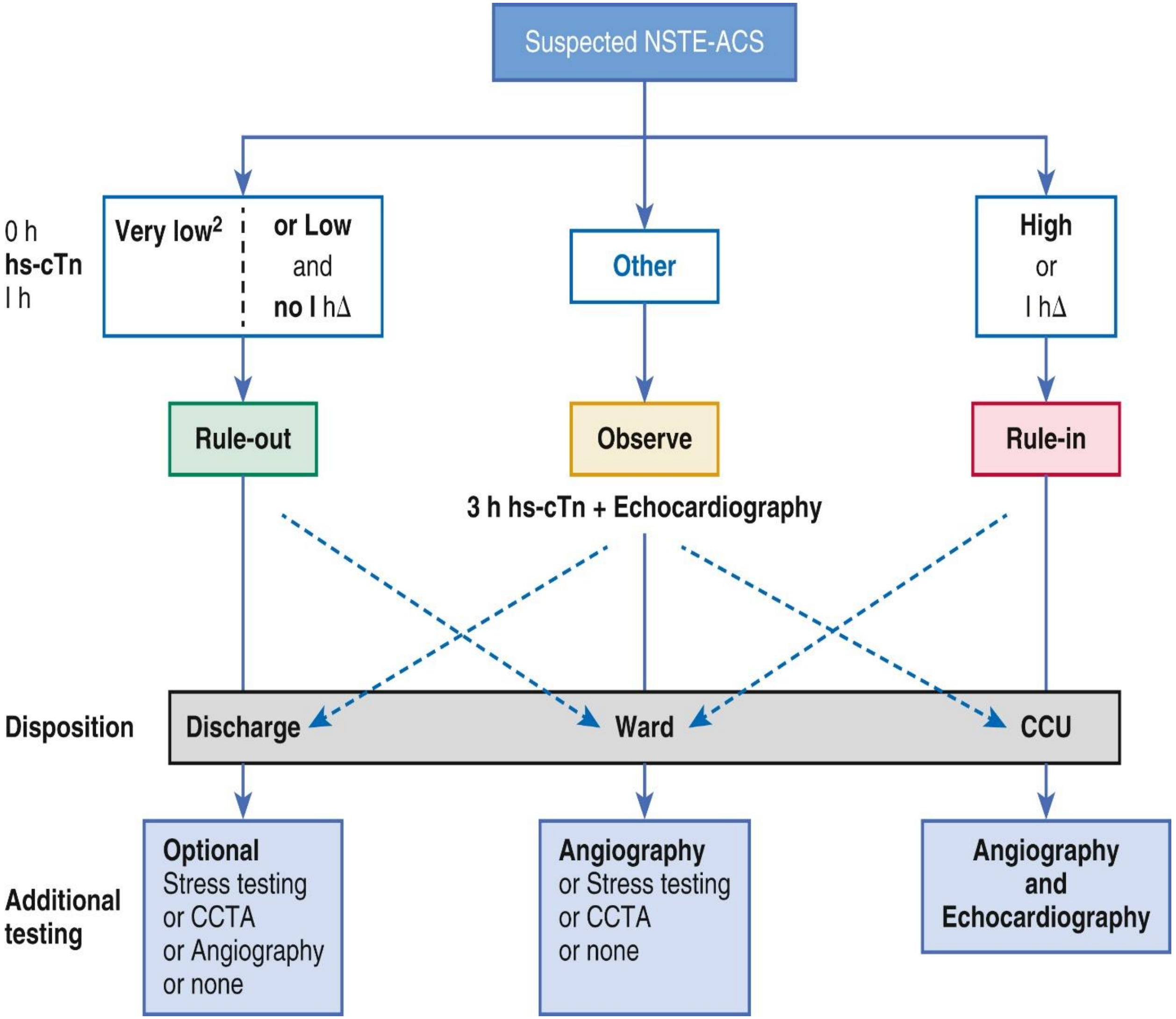
Hs-Troponin for Early Diagnosis of ACS



Risk of	Low risk	Intermediate risk	High risk
MI at index visit	<0.3%	=10%	>65%
30 day MACE	<0.5%	15-20%	>70%



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Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine 2022

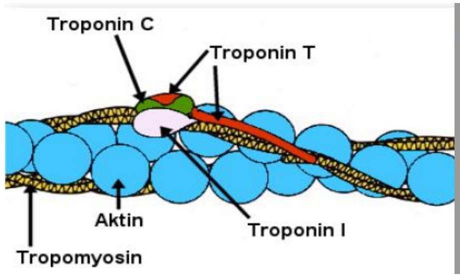
↑ Troponin

ACS
- -STEMI
- -NSTEMI

Non-coronary related
- -Myocarditis
- -Pulmonary embolism
- -Acute/chronic LV failure
- -Septic shock
- -Arrhythmias

Post procedure
- -Cardioversion
- -Catheter ablation/ Bx
- -PCI

"false" +ve in renal dysfunction

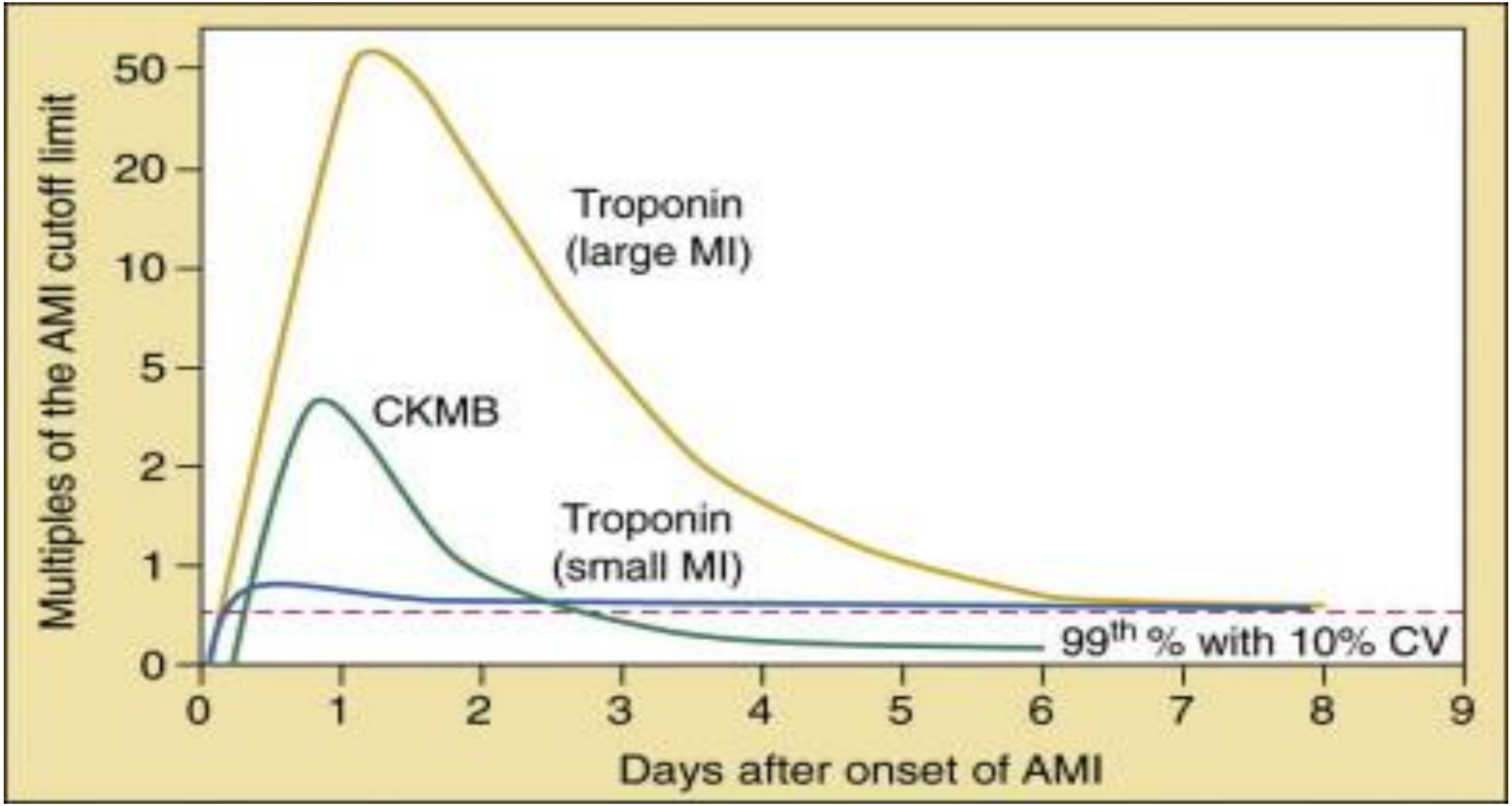


Renal insufficiency (Creatinine > 1.5 mg/dl)		
	Troponin T	15 - 20%
	Troponin I	10%
Dialysis		
	Troponin T	17 - 30%
	Troponin I	4 - 15%

• **Chronic or acute renal dysfunction**
• Severe congestive **heart failure** – acute and chronic
• **Hypertensive crisis**
• **Tachy- or bradyarrhythmias**
• **Pulmonary embolism**, severe pulmonary hypertension
• Inflammatory diseases, e.g. **myocarditis**
• Acute neurological disease, including **stroke**, or subarachnoid haemorrhage
• Aortic dissection, aortic valve disease or hypertrophic cardiomyopathy
• Cardiac contusion, ablation, pacing, cardioversion, or endomyocardial biopsy
• Hypothyroidism
• Apical ballooning syndrome (Tako-Tsubo cardiomyopathy)
• Infiltrative diseases, e.g. amyloidosis, haemochromatosis, sarcoidosis, scleroderma
• Drug toxicity, e.g. adriamycin, 5-fluorouracil, herceptin, snake venoms
• Burns, if affecting >30% of body surface area
• Rhabdomyolysis
• Critically ill patients, especially with respiratory failure, or sepsis

ESC Guideline 2011

Biomarkers for ACS



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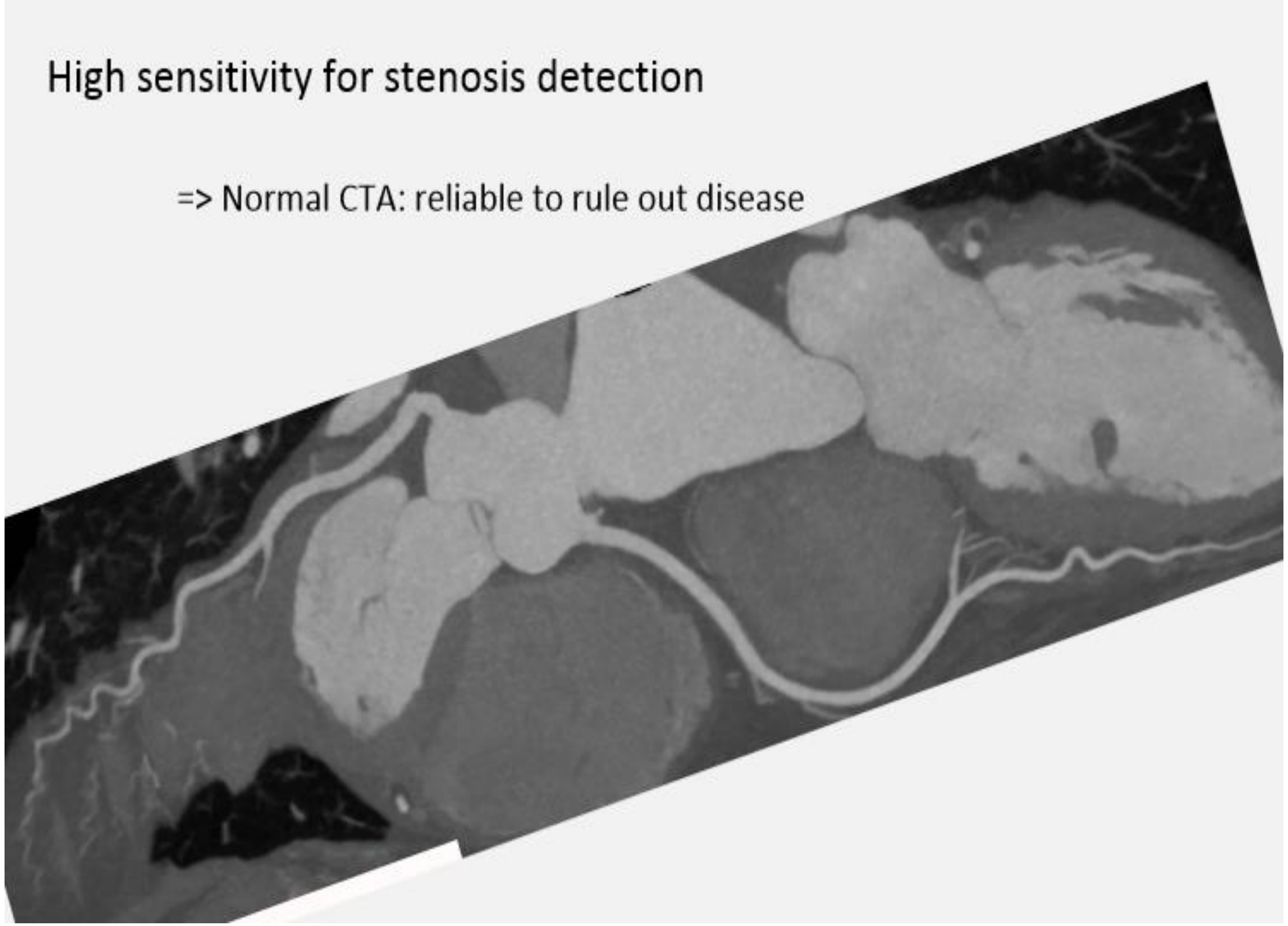
Coronary CT Angiography



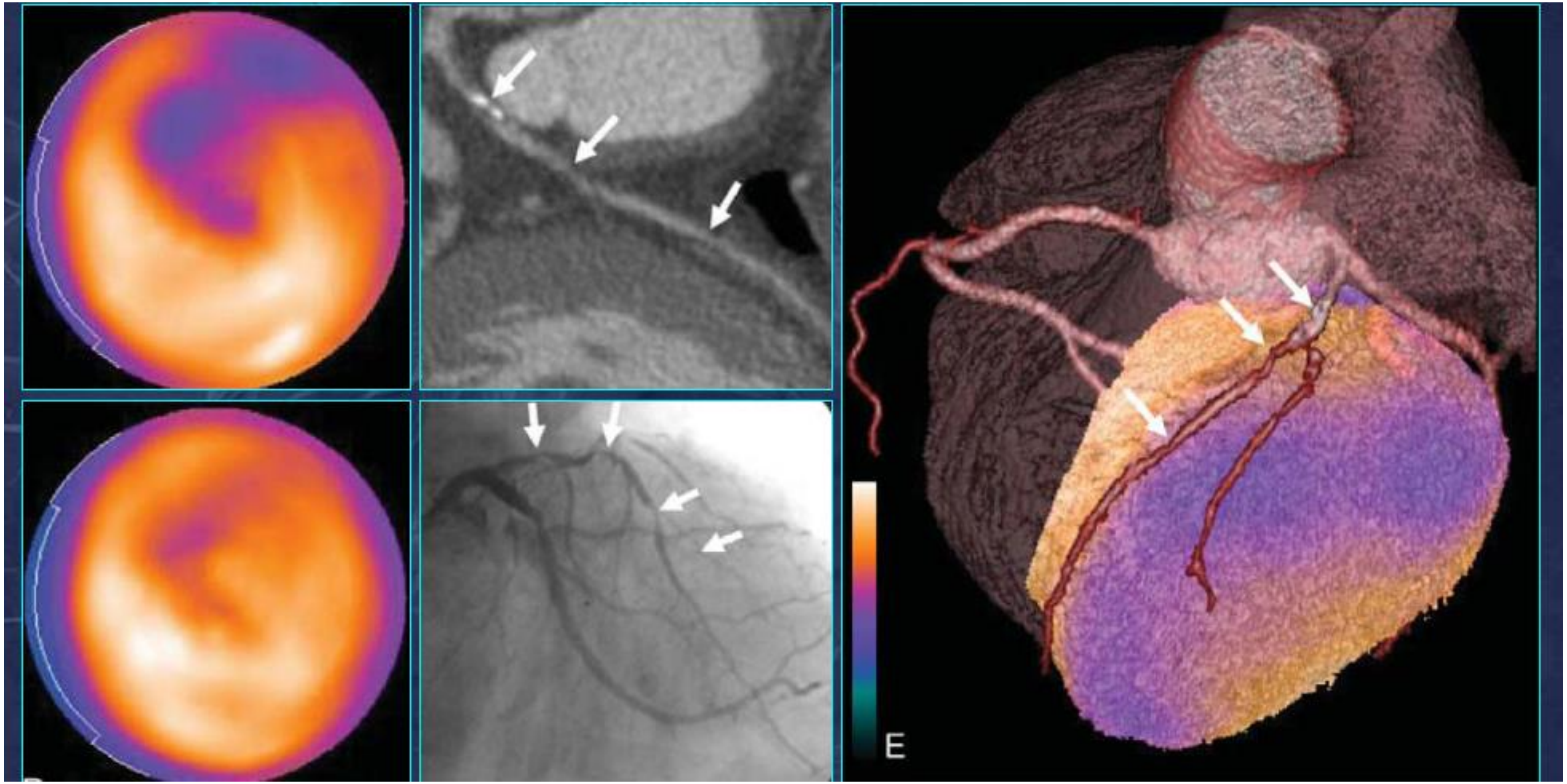
Limitations:

-Radiation, esp for young pt but reduce with ECG trigger, low HR and faster scanner

-Calcified vessels/ Previous stenting

-Rapid or irregular heart rate

Role of Non-invasive Assessment

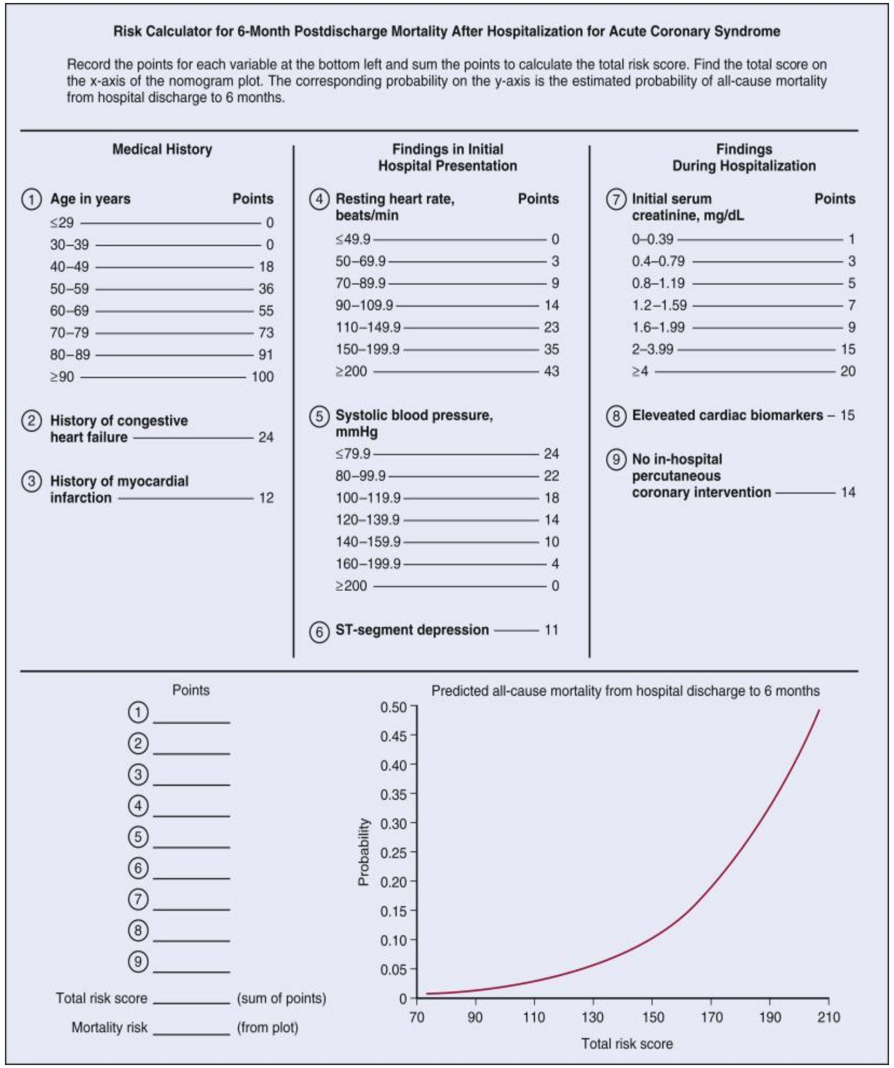


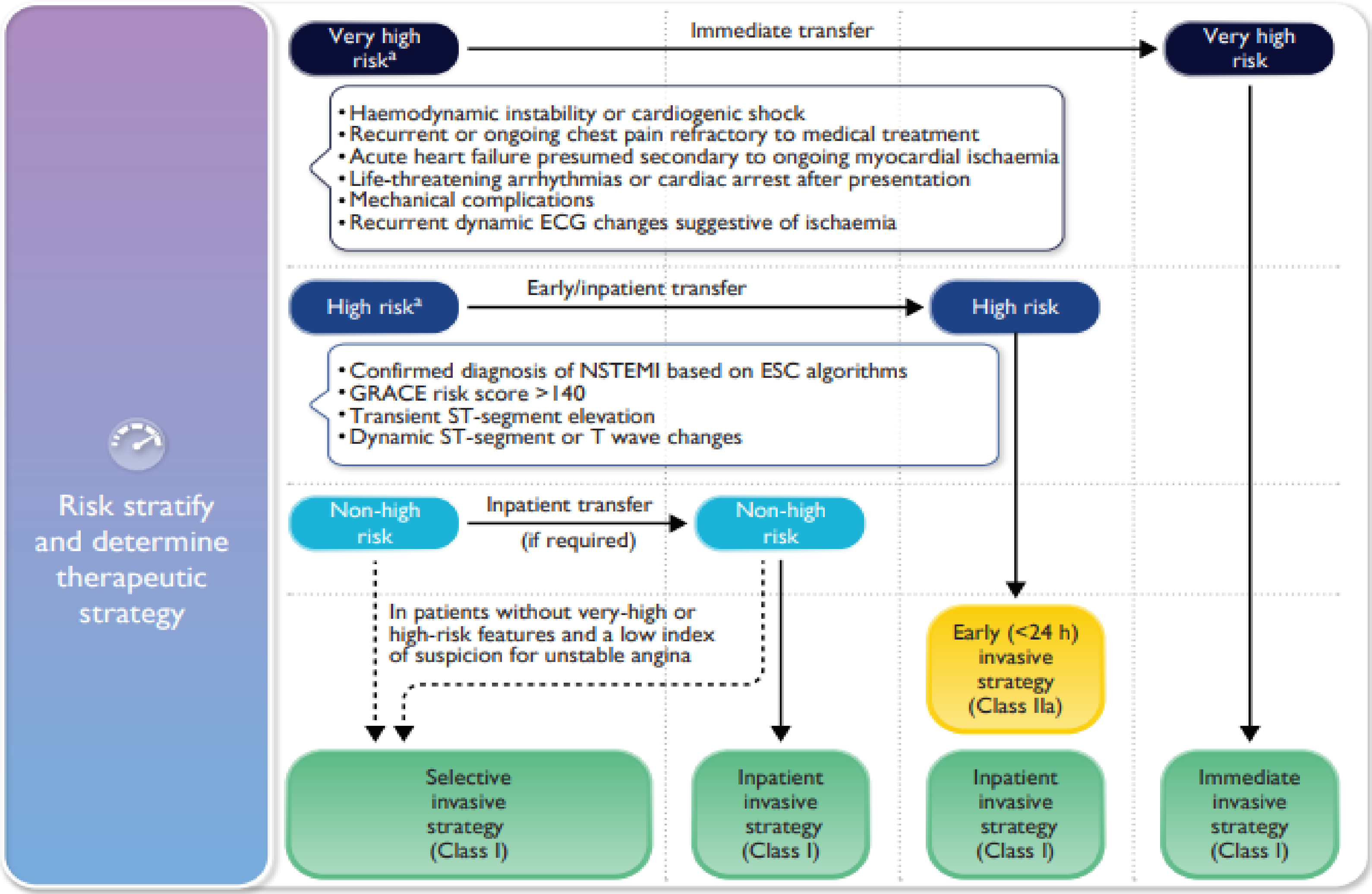
Braunwald Classification of Risk for Patients with Unstable Angina

Feature	High Risk	Intermediate Risk	Low Risk
History	Accelerating tempo of ischemic symptoms in preceding 48 hrs	Prior MI, peripheral or cerebrovascular disease, CABG, or prior aspirin use	
Character of Pain	Prolonged ongoing (>20 min) rest pain	Prolonged (>20 min) rest angina, now resolved, with moderate or high likelihood of CAD	New-onset or progressive CCS Class III or IV angina the past 2 weeks
Clinical Findings	• Pulmonary edema • New or worsening MR murmur • S_3 or new/worsening rale • Hypotension, bradycardia, tachycardia • Age >75 years	Age > 70 years	
ECG	• Angina at rest with transient ST-segment changes >0.05 mV • New or presumed new BBB • Sustained ventricular tachycardia	• T-wave inversions >0.2 mV • Pathological Q waves	Normal or unchanged ECG during an episode of chest discomfort
Cardiac Markers	Elevated (TnT or TnI >0.1 mg/mL)	Slightly elevated (TnT >0.01 but <0.1 ng/mL)	Normal

Available at: www.acc.org/clinical/guidelines/unstable/unstable.pdf.

Risk Assessment for ACS





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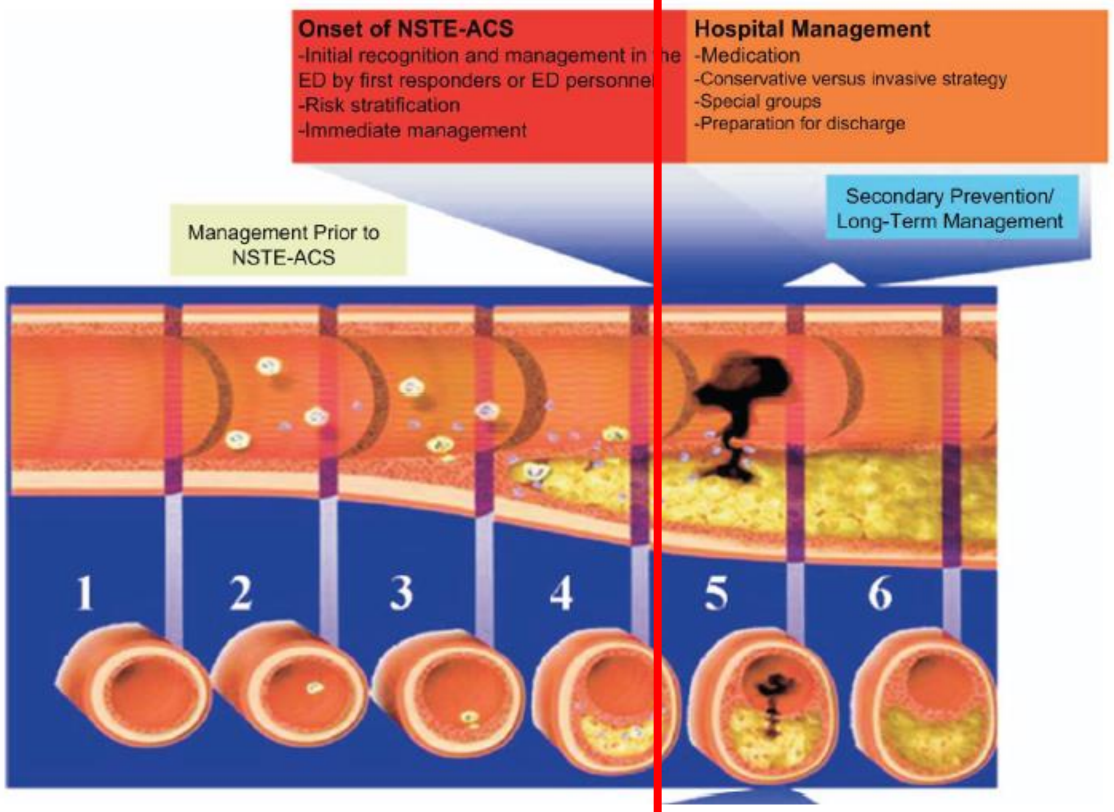
Non-ST-Segment Elevation Acute Coronary Syndrome

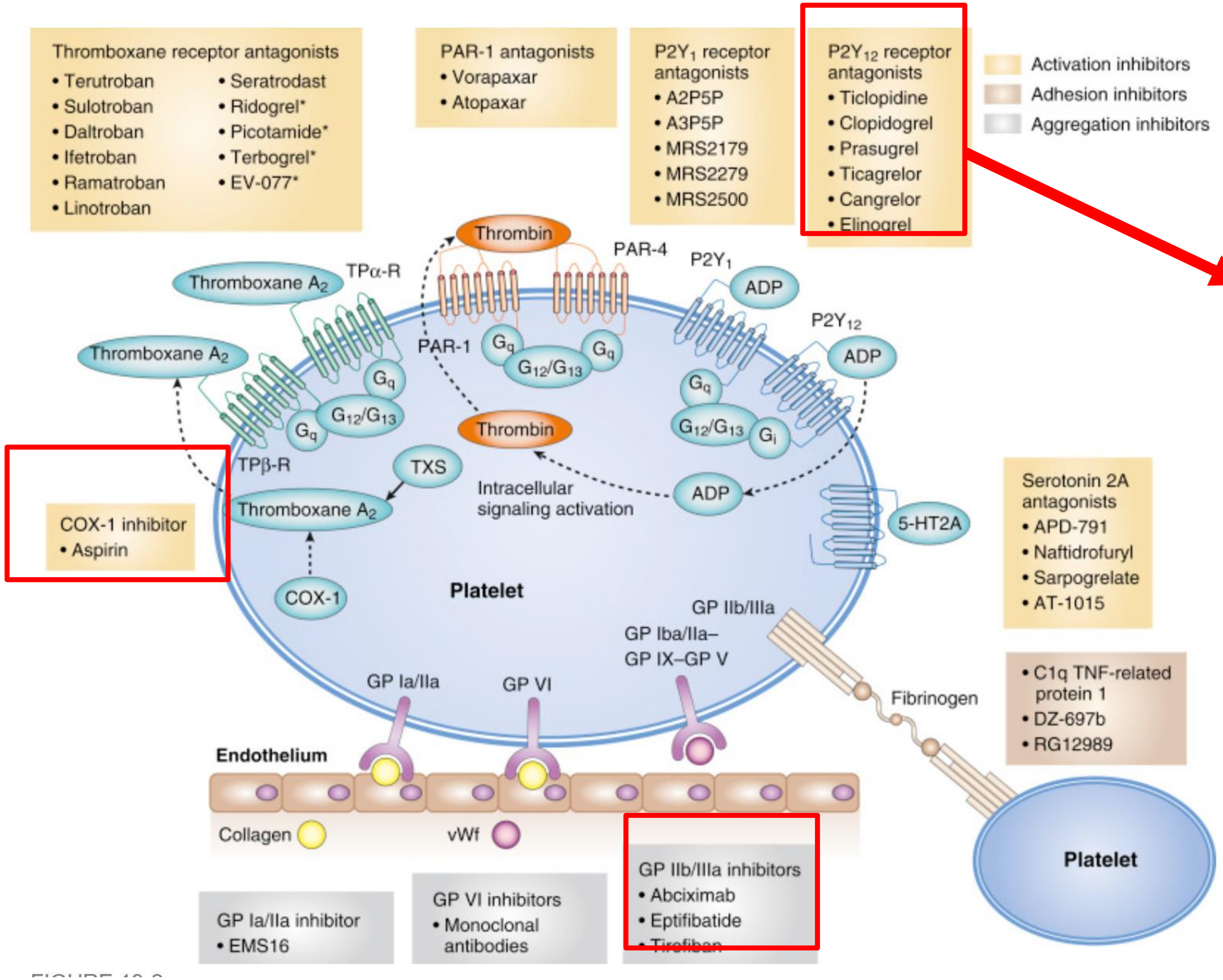
- Epidemiology
- Pathophysiology
- Diagnosis and Assessment
- **Initial Therapies and Management**
- Long-term Management

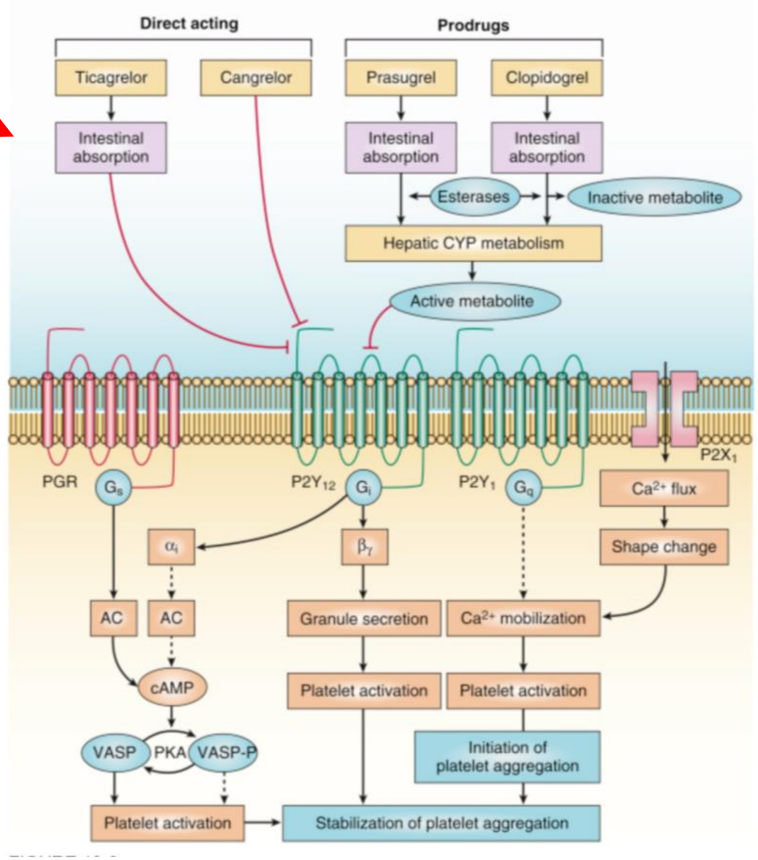
Management of NSTE-ACS

- ➢Initial Rx and anti-ischemic therapy
- ➢Antithrombotic therapy
  - \- Anti-platelet therapy
  - \- Anti-coagulant therapy
- ➢Strategy
  - \- Invasive
  - \- Conservative



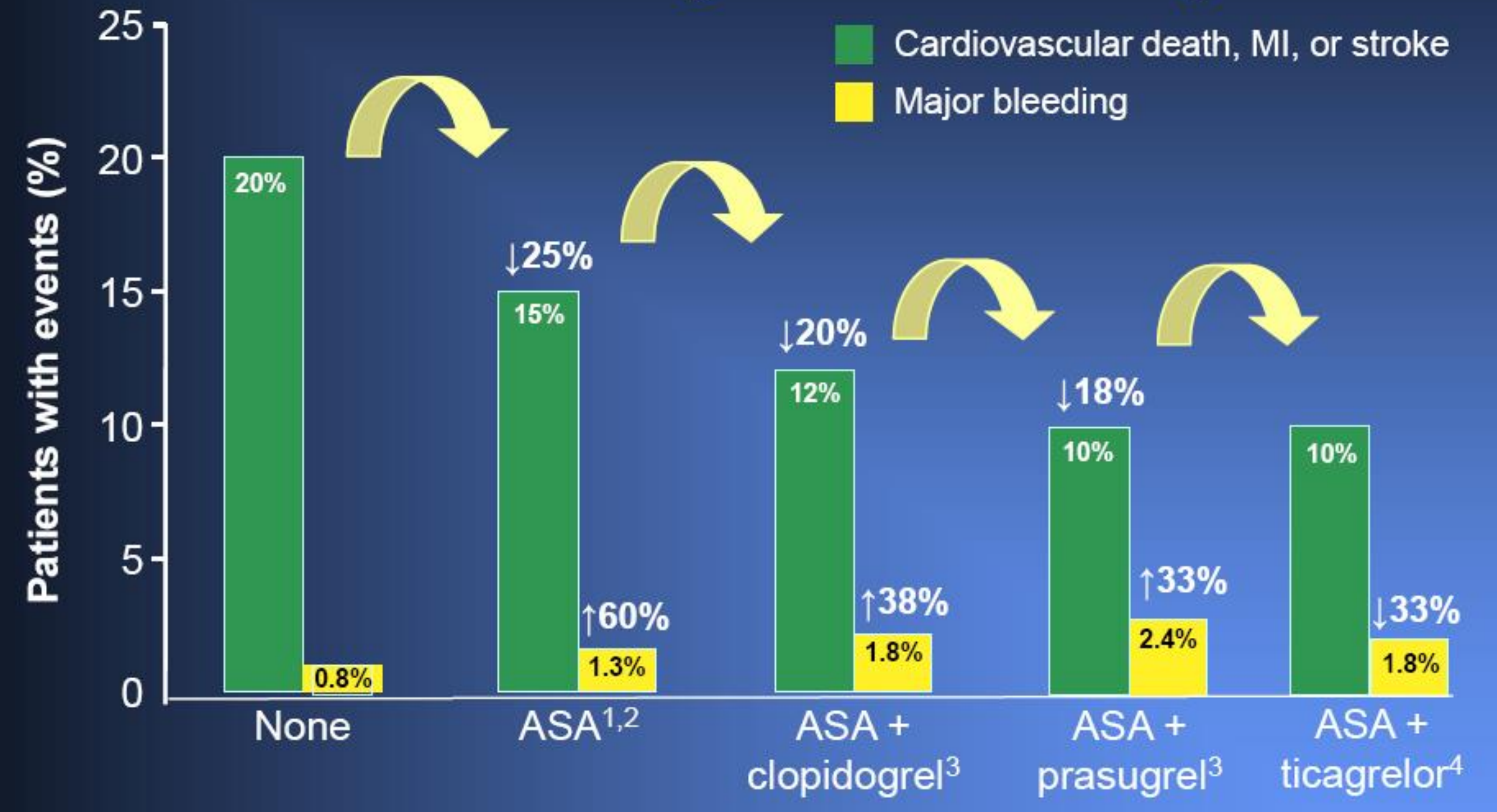
Antiplatelet Therapies for ACS





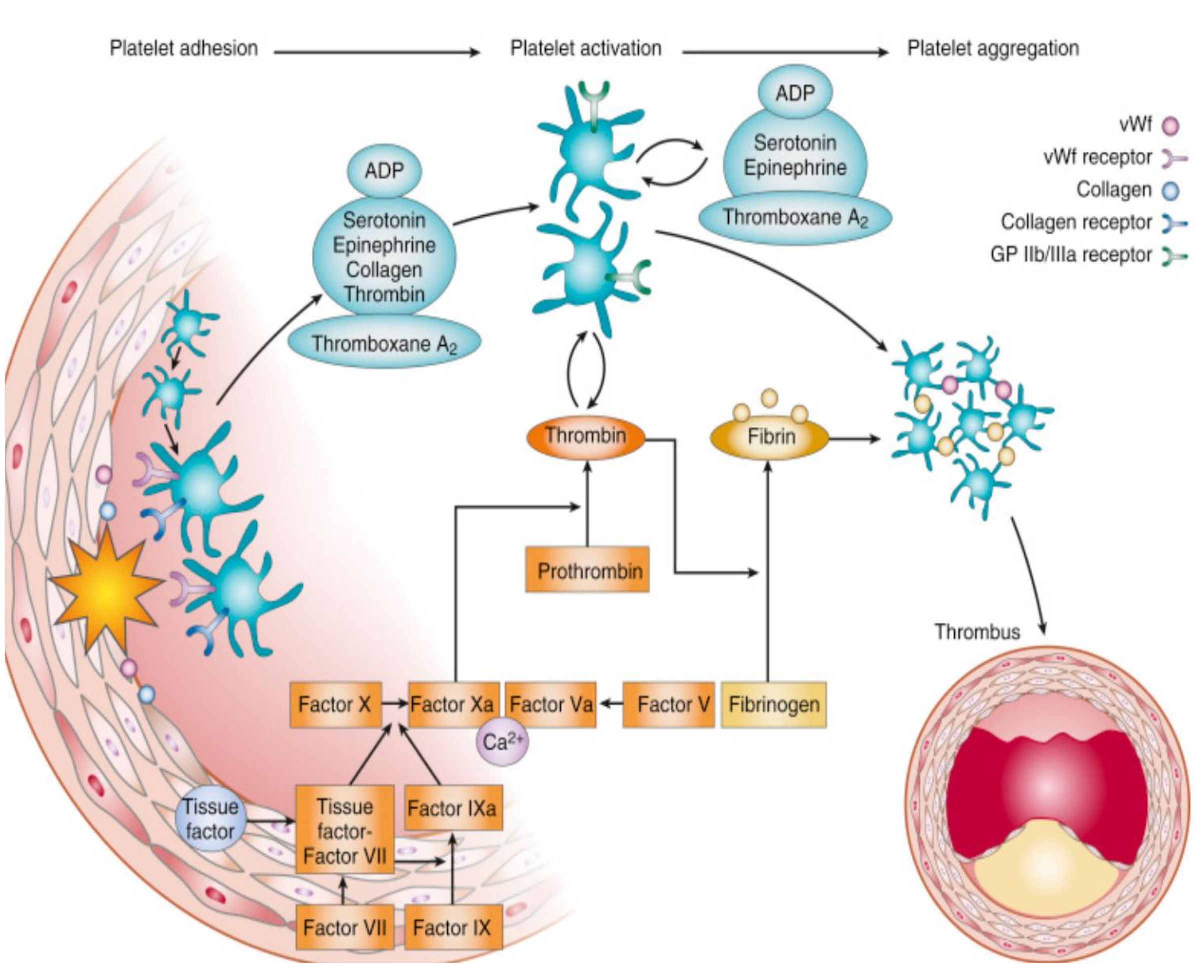
Myocardial Infarction: A Companion to Braunwald's Heart Disease

Anti-platelet Agents in ACS

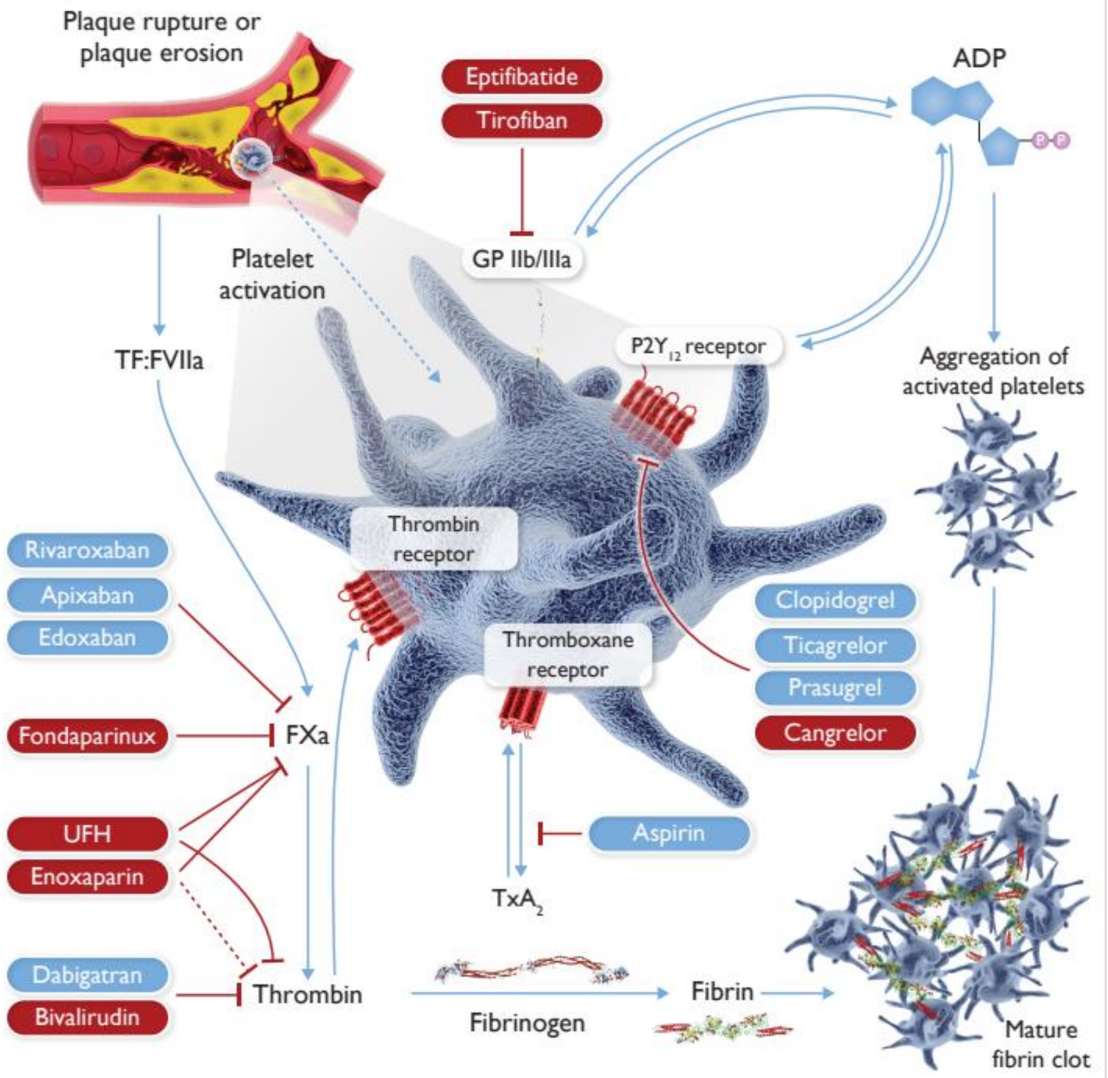


1. Antiplatelet Trialists' Collaboration. *BMJ* 1994;308:81–106; 2. Antithrombotic Trialists' Collaboration. *BMJ* 2002; 324:71–86; 3. Wiviott *et al. N Engl J Med* 2007; 357:2001–2015. 4. PLATO *N Engl J Med* 2009;361:1045-1057

Targets for Anti-thrombotic Rx for ACS



Myocardial Infarction: A Companion to Braunwald's Heart Disease



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Antiplatelet Therapies ($P2Y_{12}$Inhibitor) for ACS

	Clopidogrel	Prasugrel	Ticagrelor	Cangrelor
Pharmacological class	Thienopyridine	Thienopyridine	CPTP	ATP analogue
Receptor blockade	Irreversible	Irreversible	Reversible	Reversible
Administration route	Oral	Oral	Oral	IV
Frequency	Once daily	Once daily	Twice daily	Bolus plus infusion
Prodrug	Yes	Yes	No *	No
Onset of action	2–8 hrs	30 min–4 hrs †	30 min–4 hrs †	2 min
Offset of action	7–10 days	7–10 days	3–5 days	30–60 min
CYP drug interaction	CYP2C19	No	CYP3A	No
Approved settings	ACS and stable CAD PCI	ACS undergoing PCI	ACS (full spectrum)	$P2Y_{12}$ receptor inhibitors naïve patients undergoing PCI

Myocardial Infarction: A Companion to Braunwald's Heart Disease

Antiplatelet Therapies for ACS

First medical contact
NSTEMI

Pretreatment	
Preferred agent	**Alternative agents**
None established*	**Ticagrelor** **Clopidogrel** (if ticagrelor is not available or contraindicated)

Decision on management

Medical management	
Preferred agent	**Alternative agents**
Ticagrelor	**Clopidogrel** (if ticagrelor is not available or contraindicated)

Invasive management	
Preferred agent	**Alternative agents**
If PCI performed: **Prasugrel** (after defining coronary anatomy) or **ticagrelor** (choice should be based taking into consideration contraindications and precautions).† Consider **cangrelor** in patients not pretreated. **If CABG needed:** withdraw ticagrelor for 5 days and prasugrel for 7 days. **If no revascularization:** see medical management.	**If PCI performed:** **Clopidogrel** (if ticagrelor and prasugrel are not available or contraindicated) **If CABG needed:** withdraw clopidogrel for 5 days. **If no revascularization:** see medical managment.

Discharge	
Preferred agent	**Alternative agents**
Ticagrelor (if patient was previously treated with clopidogrel, switch to ticagrelor)	**Clopidogrel** (if ticagrelor is not available or contraindicated)

Discharge	
Preferred agent	**Alternative agents**
If PCI performed: continue **prasugrel** or **ticagrelor** (choice should be based taking into consideration contraindications and precautions).† **If CABG performed:** resume ticagrelor as soon as possible. **If no revascularization:** see medical managment.	**Clopidogrel** (if prasugrel and ticagrelor are not available or contraindicated) **If CABG performed:** resume clopidogrel as soon as possible. **If no revascularization:** see medical managment.

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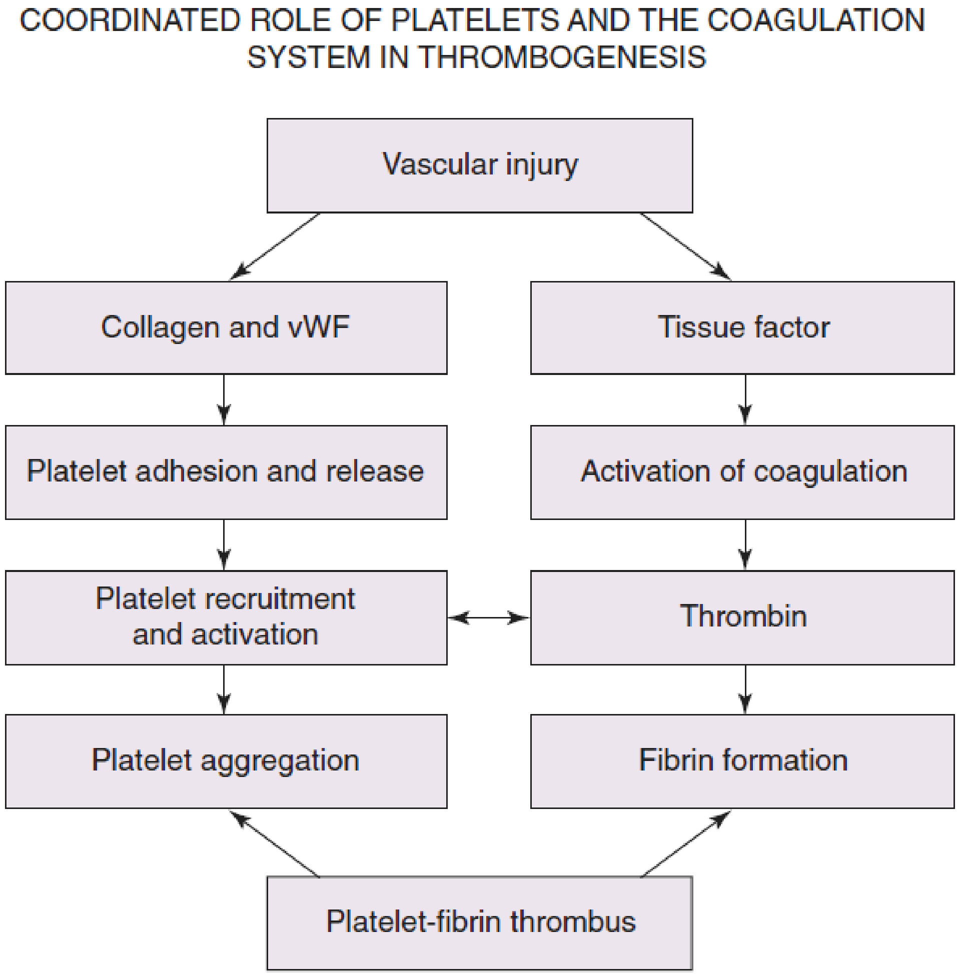
Myocardial Infarction: A Companion to Braunwald's Heart Disease

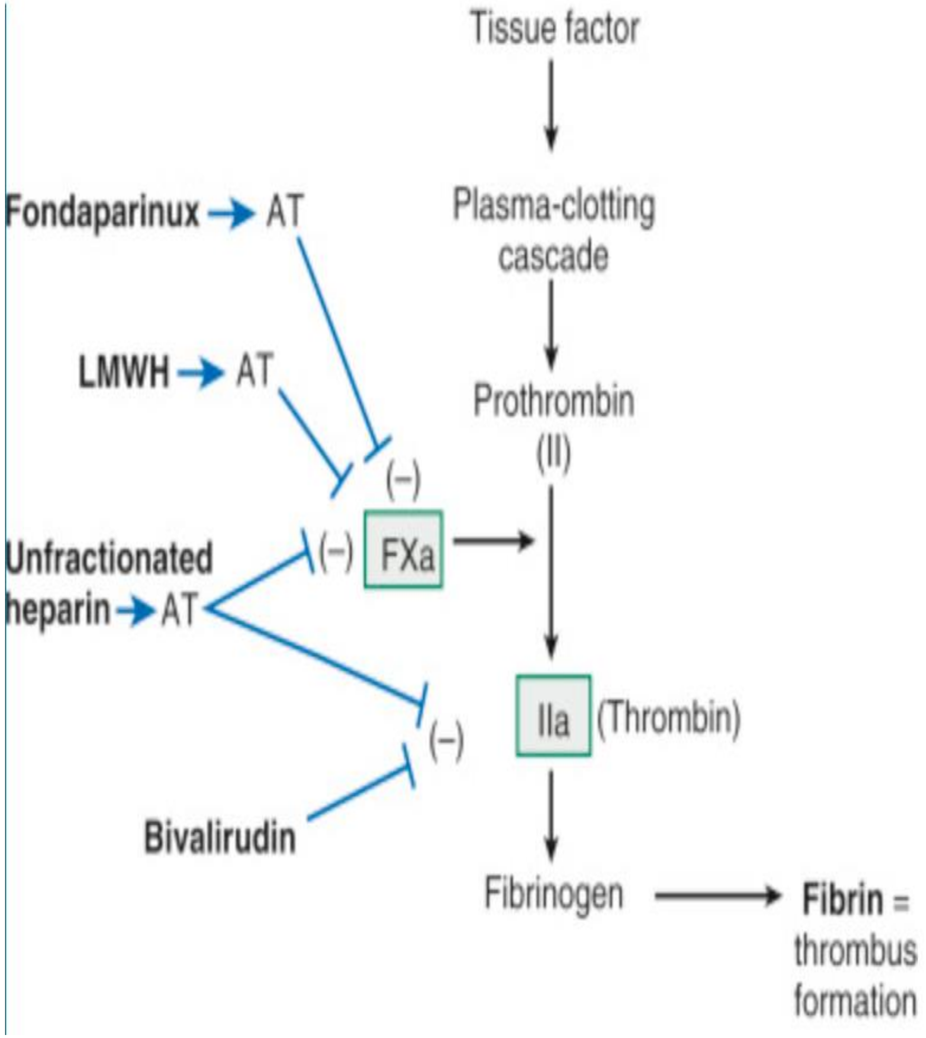
Antiplatelet Therapies (GP IIb3a Inhibitors) for ACS

	Abciximab	Eptifibatide	Tirofiban
Trade name	ReoPro	Integrilin	Aggrastat
Molecule	Fragment antigen binding (Fab) 7E3	Synthetic peptide	Nonpeptide mimetic
Molecular weight	~50,000	~800	~500
Stoichiometry (drug to GP IIb/IIIa)	~1.5:1	>>100:1	>>100:1
Binding	Noncompetitive	Competitive	Competitive
Half-life	Plasma: 10–15 hrs Biologic: 12–24 hrs	Plasma: 2–2.5 hrs Biologic = plasma	Plasma: 2–2.5 hrs Biologic = plasma
PCI dosing	Bolus: 0.25 mg/kg (10–60 min) Infusion: 0.125 μg/kg/min (12h)	Bolus: 180 μg/kg * + 180 μg/kg (after 10 min) Infusion: 2 μg/kg/min (24–48 hrs) †	Bolus: 25 μg/kg (30 min) Infusion: 0.10 μg/kg/min (48 hrs)
Renal adjustment	No	Bolus: 180 μg/kg Infusion: 1 μg/kg/min (24–48 hrs)	Bolus: 12.5 μg/kg (30 min) Infusion: 0.10 μg/kg/min (48 hrs)

Myocardial Infarction: A Companion to Braunwald's Heart Disease

Anticoagulant





Myocardial Infarction: A Companion to Braunwald's Heart Disease
Braunwald's Heart Disease, 2020

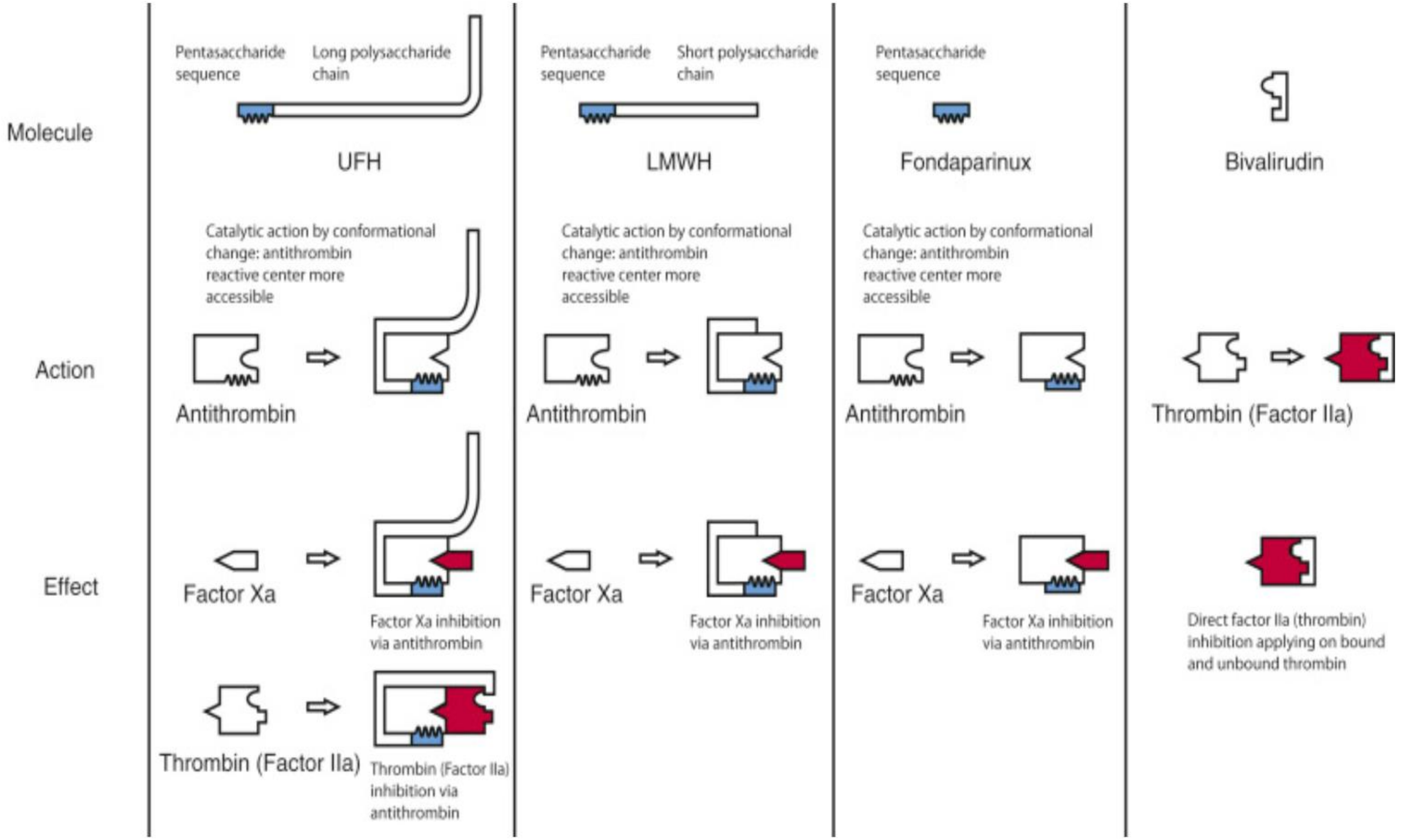
Anticoagulants for ACS

Medication	UFH	Enoxaparin	Fondaparinux	Bivalirudin
Mechanism	AT mediated factor Xa and thrombin inhibitor	AT mediated factor Xa and thrombin inhibitor	Indirect factor Xa inhibitor	Direct thrombin inhibitor, reversible
Route of administration	IV–SC	IV–SC	SC	IV
Half-life	1–2 h	5–7 h	17–21 h	25 min
Molecular weight	3–30 kDa	2–10 kDa	1.7 kDa	2.2 kDa
Metabolism	Hepatic	Hepatic	Excreted largely as unchanged drug	Plasma proteases
Elimination	Extra-renal	Renal	Renal	Renal
Onset	Immediate (IV)	Immediate (IV) 3–5 h (SC)	2–3 h (SC)	Immediate
Notes	Inactive on clot-associated thrombin	Inactive on clot-associated thrombin More factor sXa selectivity	Inactive on clot-associated thrombin	Inhibits clot-associated thrombin
Incidence of HIT	1%–3%	≤0.2%	Negligible	None
Antidote	Protamine	Protamine (partial)	None	None

Anticoagulants for ACS

TABLE 22–1 Advantages of Low-Molecular-Weight Heparin over Unfractionated Heparin

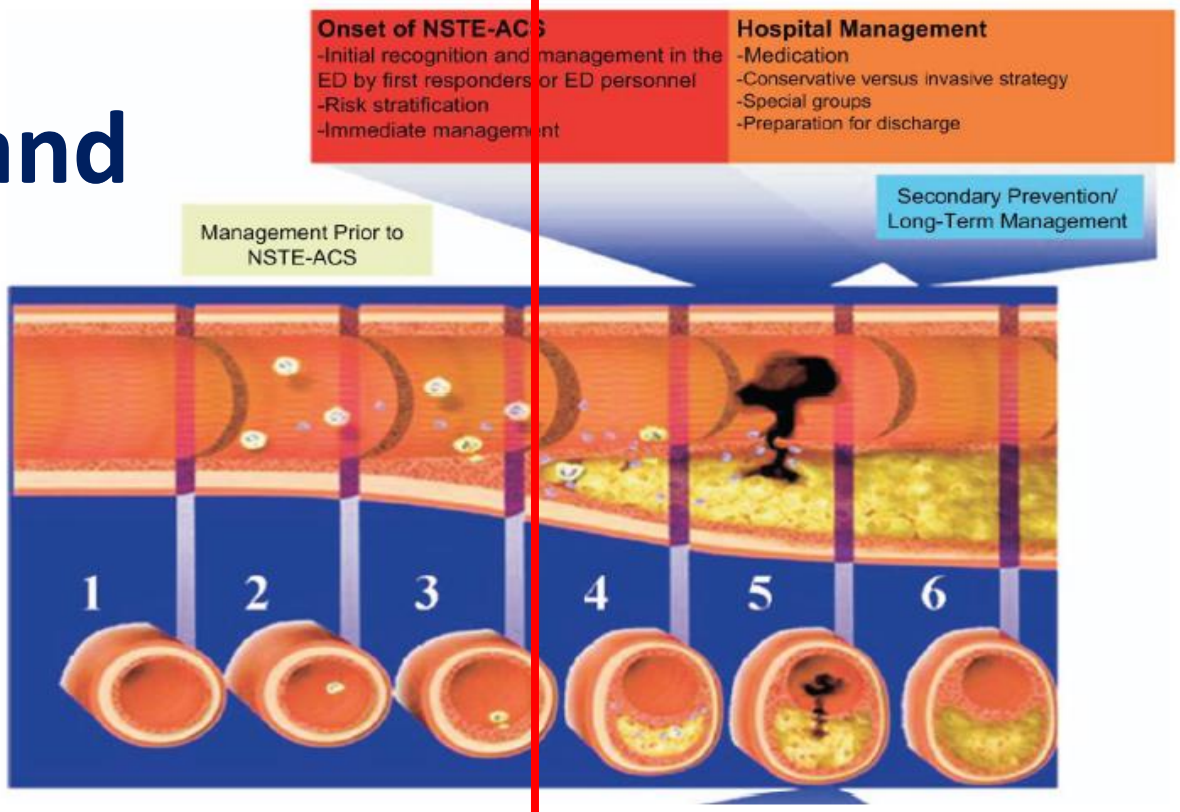
Advantage	Effect	Clinical Consequence
Better bioavailability	Higher drug levels achieved after subcutaneous injection	Subcutaneous administration for prophylaxis or treatment
Reduced binding to proteins	More predictable anticoagulant response	Routine monitoring is unnecessary
Reduced binding to and clearance by endothelial cells and macrophages	Clearance predominantly by renal mechanisms	Longer plasma half-life permits once-daily dosing
Reduced binding to osteoblasts	Reduced activation of osteoclasts	Lower incidence of osteopenia and less risk of heparin-associated osteoporosis and fracture with prolonged treatment
Reduced affinity for platelets and platelet factor 4	Less platelet activation with reduced release of platelet factor 4; reduced formation of complexes of heparin/platelet factor 4	Reduced incidence of heparin-induced thrombocytopenia



Myocardial Infarction: A Companion to Braunwald's Heart Disease

Aims of Treatment for NSTE-ACS

- ➢Relieve symptoms:
  - Anti-ischemic therapy
- ➢Prevention progression and complications
  - Anti-thrombotic Rx
  - Revascularization



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Primary anti-ischemic therapeutic measures

General	Bed rest and ECG/BP monitoring
Oxygen	Insufflation (4 to 8 L/min) if oxygen saturation is <90%
Nitrates	Sublingually or intravenously (caution if systolic blood pressure <90 mmHg)
Morphine	3 to 5mg intravenous or subcutaneous depending on pain seventy
Oral beta-blocker or CCB	Particularity, if tachyardia or hypertension without sign of heart failure Diltiazem or verapamil if BB failed or contraindicated
Atropine	0.5-1 mg intravenously, if bradycardia or vagal reaction

Contraindication to Anticoagulant

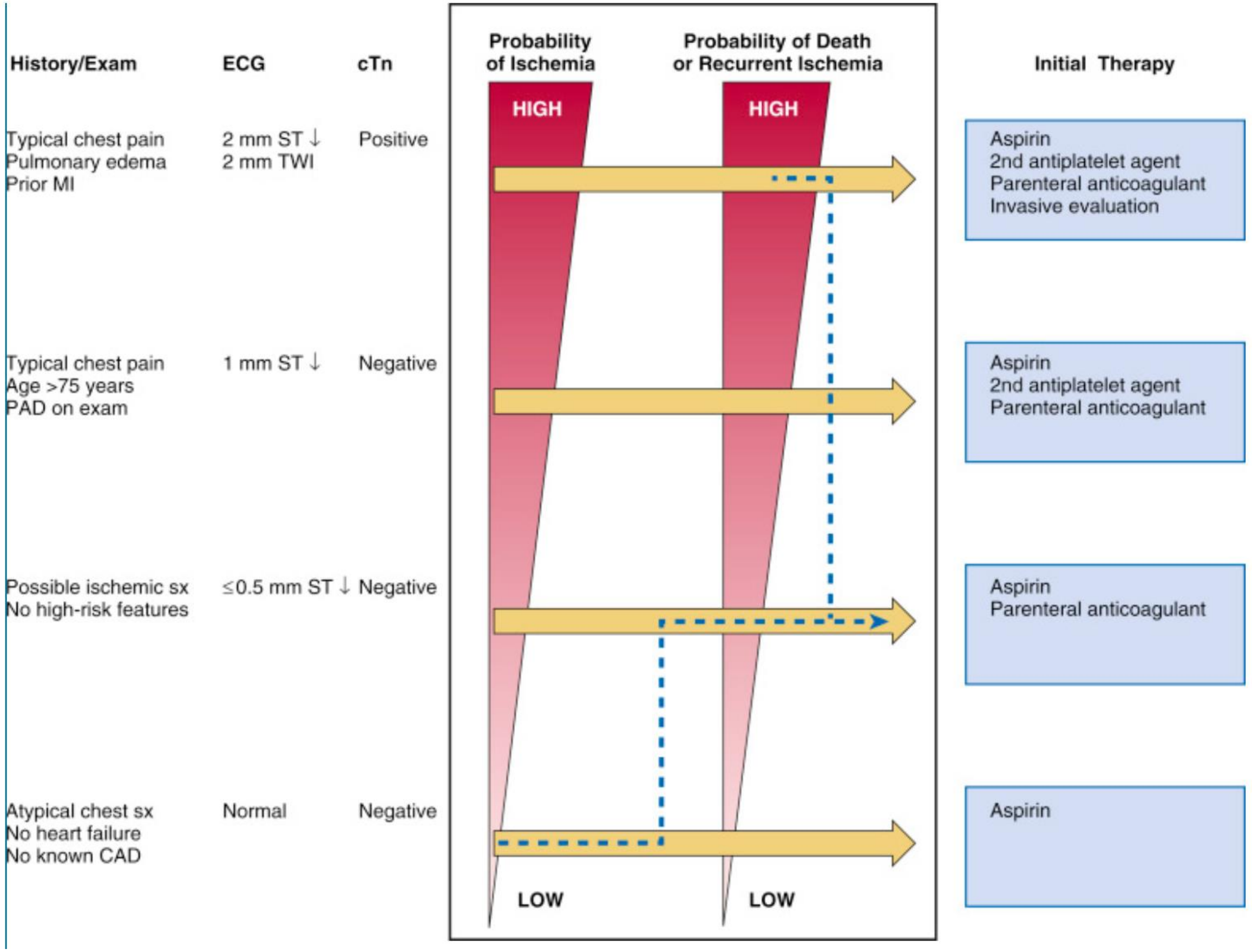
Absolute

- Active bleeding
- Severe bleeding diathesis
- Severe thrombocytopenia
- Recent neurosurgery, ocular surgery (excluding cataract surgery), or intracranial bleed

Relative

- Moderate thrombocytopenia
- Bleeding diathesis
- Brain metastases
- Recent major trauma
- Recent major abdominal surgery (<1 or 2 days)
- Gastrointestinal or genitourinary bleeding within the past 14 days
- Endocarditis
- Severe hypertension
- Systolic blood pressure > 200 mm Hg and/or diastolic blood pressure > 120 mm Hg at presentation

ACS Antithrombotic Therapies Summary



Aspirin +
SC LMWH or IV Heparin+
Clopidogrel or Ticagrelor
Prasugrel +/-
Abciximab/Tirofiban/Eptifibatide (PCI)

Aspirin +
SC LMWH or IV Heparin+
Clopidogrel or Ticagrelor
Or Prasugrel (PCI planned)

Aspirin +
SC LMWH or IV Heparin
Or Fondaparinux
Or Bivalirudin (PCI planned)

Aspirin
(loading 150-300mg then 75-100mg)

Myocardial Infarction: A Companion to Braunwald's Heart Disease

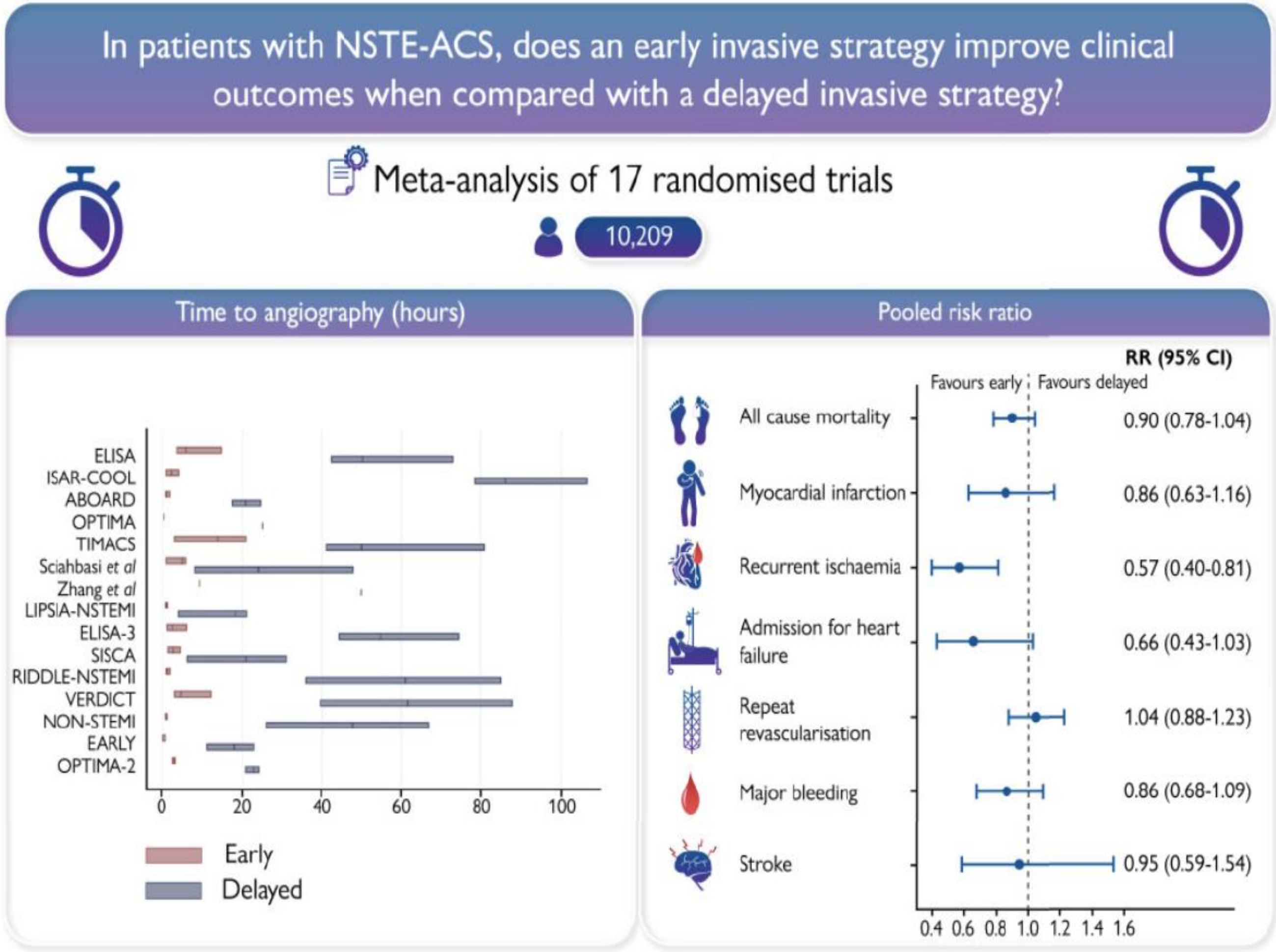
CTM, M/51 (4)

- Treated as NSTEMI ASC
- Started

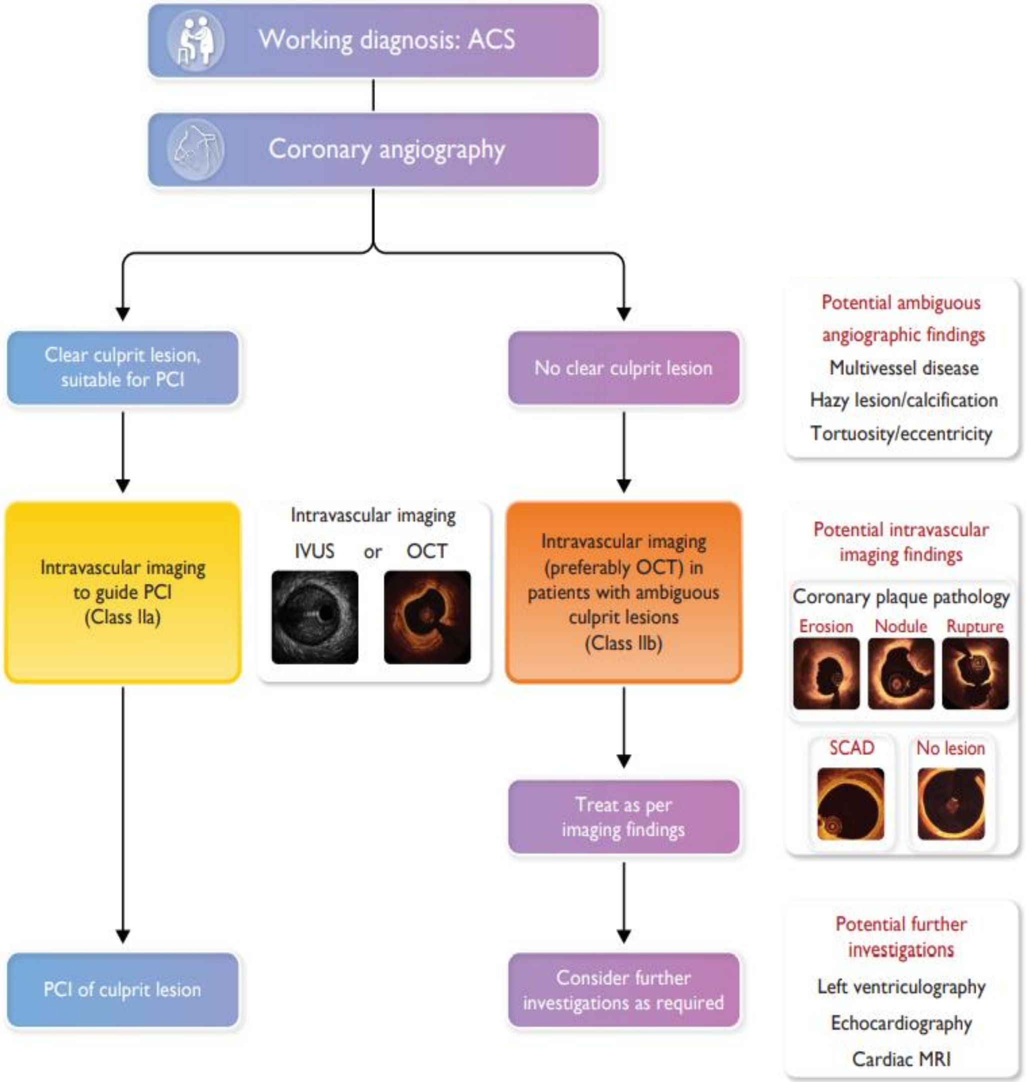
Aspirin 300mg stat then 100 daily
Enoxaparin 80mg Q12 hr
Ticagrelor 180mg loading then 90mgBD
Pantoloc 40mg daily
Rosuvastatin 20mg daily

- Echo showed impaired LVEF 40%
- Arrange for coronary angiogram +/- revascularization

Coronary Revascularization in ACS



Kite TA, et al. Eur Heart J 2022;43:3148–3160



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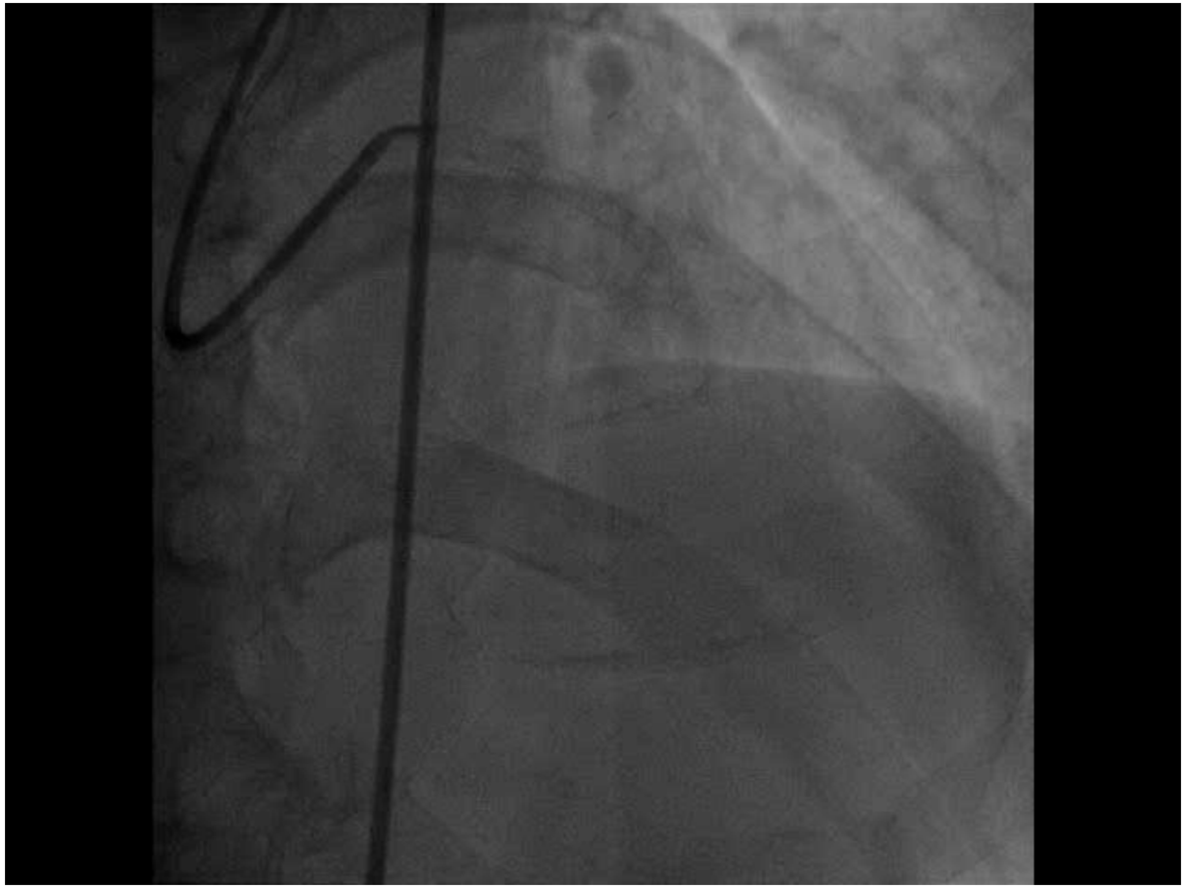
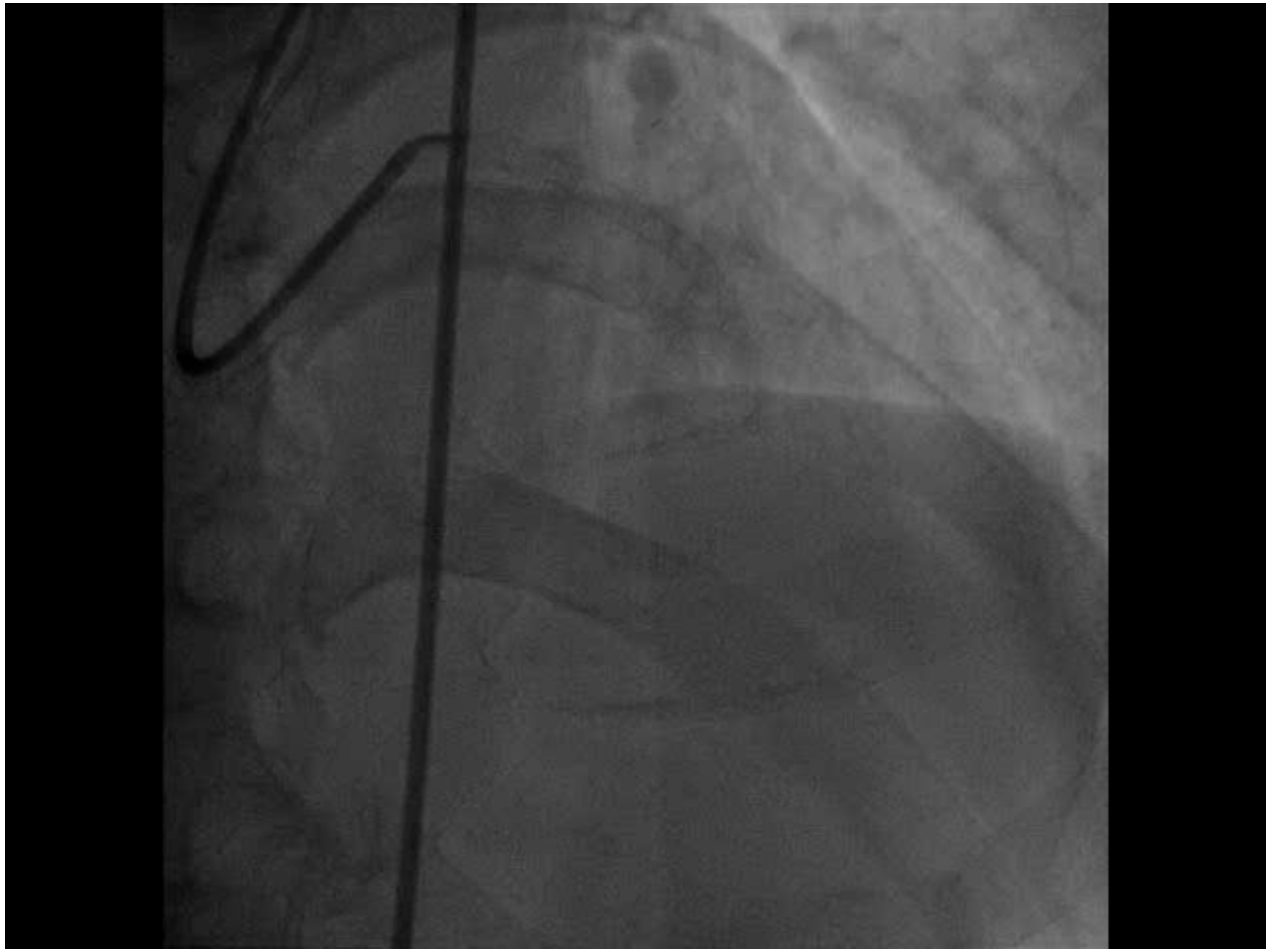
CTM, M/51 (5)



CTM, M/51 (6)



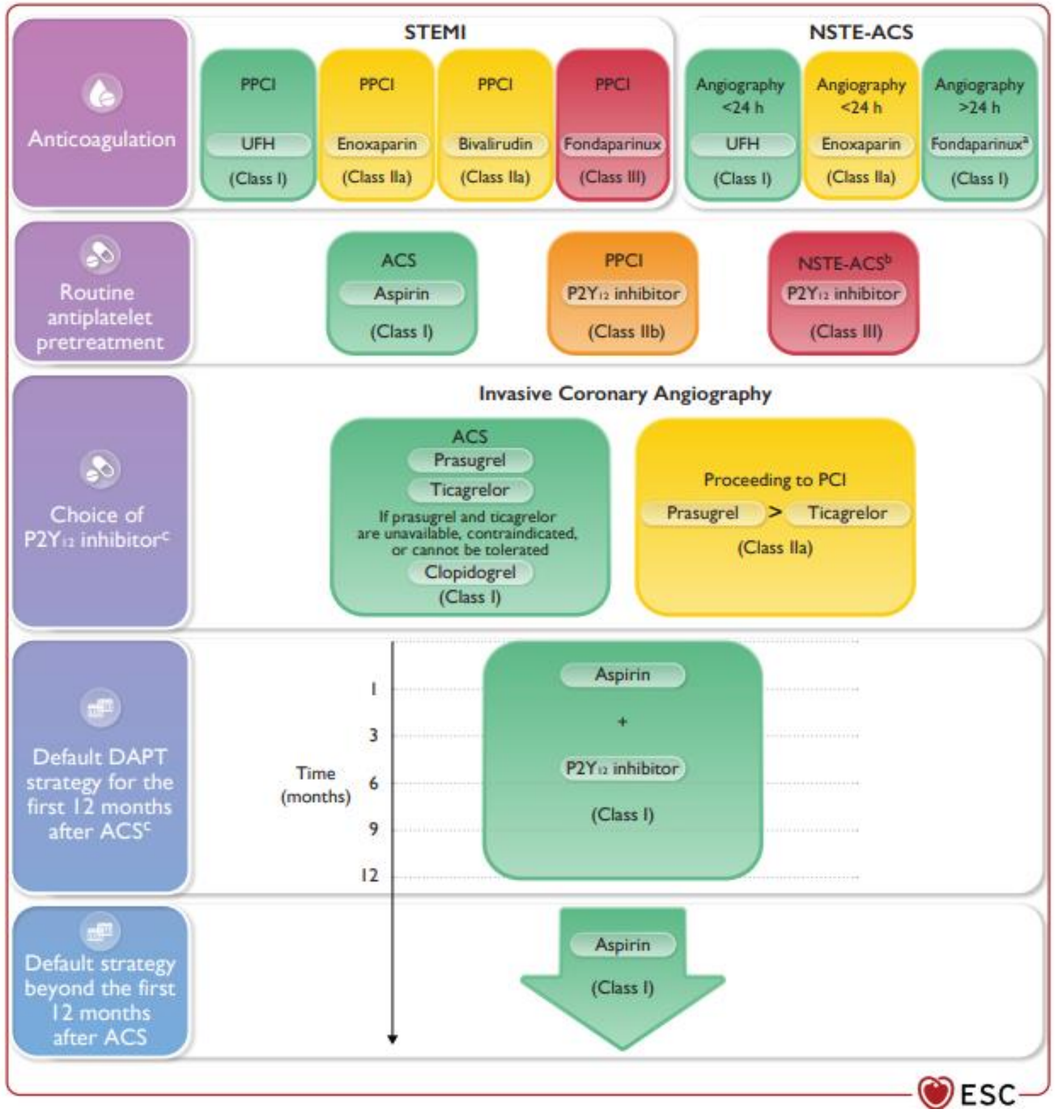
Non-ST-Segment Elevation Acute Coronary Syndrome

- Epidemiology
- Pathophysiology
- Diagnosis and Assessment
- Initial Therapies and Management
- **Long-term Management**

Secondary prevention

Anti-platelet Therapy:

- Aspirin is recommended for all patients without contraindications at dose of 75-100 mg daily
- A P2Y12 receptor inhibitor is recommended in addition to aspirin, and maintained over 12 months unless there are contraindications or an excessive risk of bleeding:
  - Ticagrelor 90 mg b.i.d.
  - Clopidogrel 75 mg/d
- Clopidogrel 75 mg qD when ASA is not tolerated because of hypersensitivity or GI intolerance



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Secondary prevention

Anti-ischemic Therapy:

- **Beta blockers unless contraindicated**
- **ACEI for patients with CHF, LV dysfunction (EF <40%), hypertension, or diabetes**
- **Calcium antagonists (diltiazem or verapamil) if contraindications to beta-blockers and no heart failure**
- **Nitrates (long-acting or short acting as prn) in the presence of angina**

Anti-ischemic Therapies for ACS

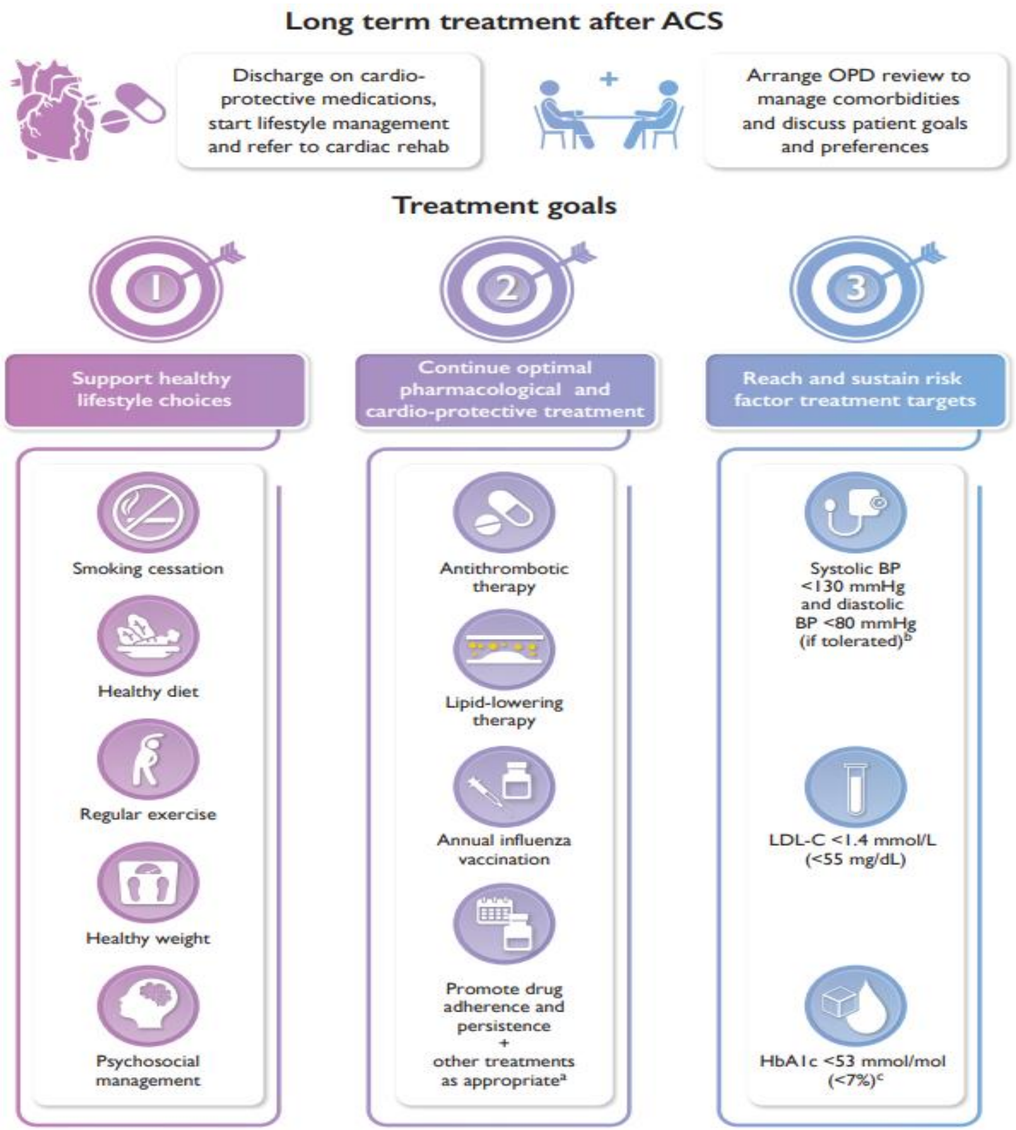
Class Of Medication	Mechanism of Action	Clinical Effects In Nste-Acs
Traditional Therapies		
Beta blockers	Decrease heart rate, blood pressure, and contractility through antagonism of beta $_1$ receptors	Decrease mortality [51]
Nitrates	Decrease preload through venodilation; vasodilate coronary arteries	No benefit on mortality
Calcium channel blockers	May vasodilate, reduce heart rate, or decrease contractility depending on specific drug	No clear benefit on mortality or reinfarction Increased reinfarction rate when short-acting nifedipine is used alone
Newer and Experimental Therapies		
Ranolazine	Inhibits late inward sodium current	Decreases recurrent ischemia and arrhythmias
Trimetazidine	Shifts myocardial metabolism from fatty acid to glucose use	Decreases short-term mortality
Nicorandil	Activates ATP-sensitive K^+ channels and dilates arterioles; may have ischemic precondition-like effect	Decreases arrhythmias and transient ischemia

Myocardial Infarction: A Companion to Braunwald's Heart Disease

Secondary prevention

Risk Factors Control:

- Stop smoking
- Optimal glycaemic control in diabetic patients (Hba1c 7%)
- Blood pressure control in hypertensive patients
- Aggressive lipid lowering with statin (LDL 1.4-1.8)
- Diet and wt control



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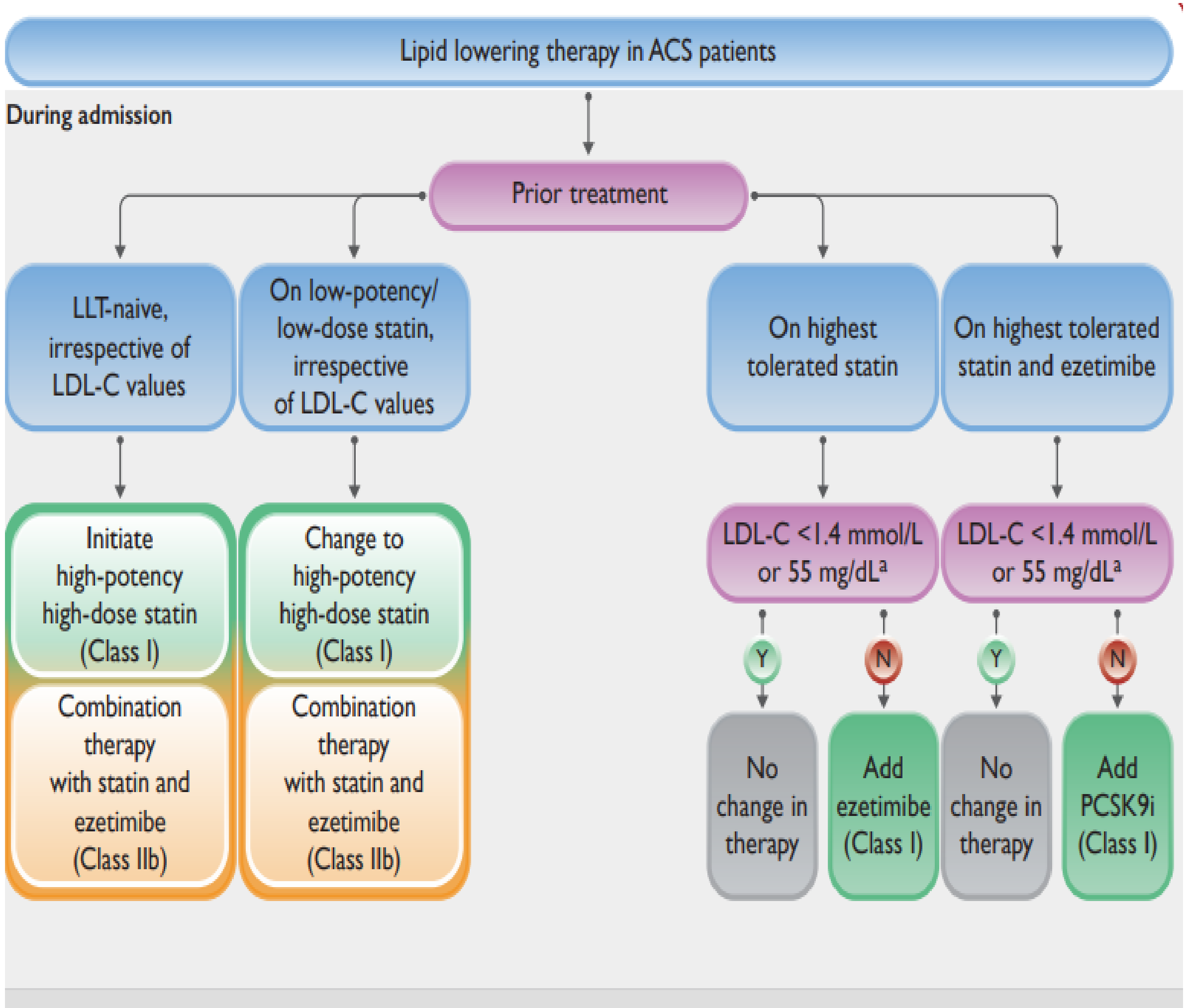
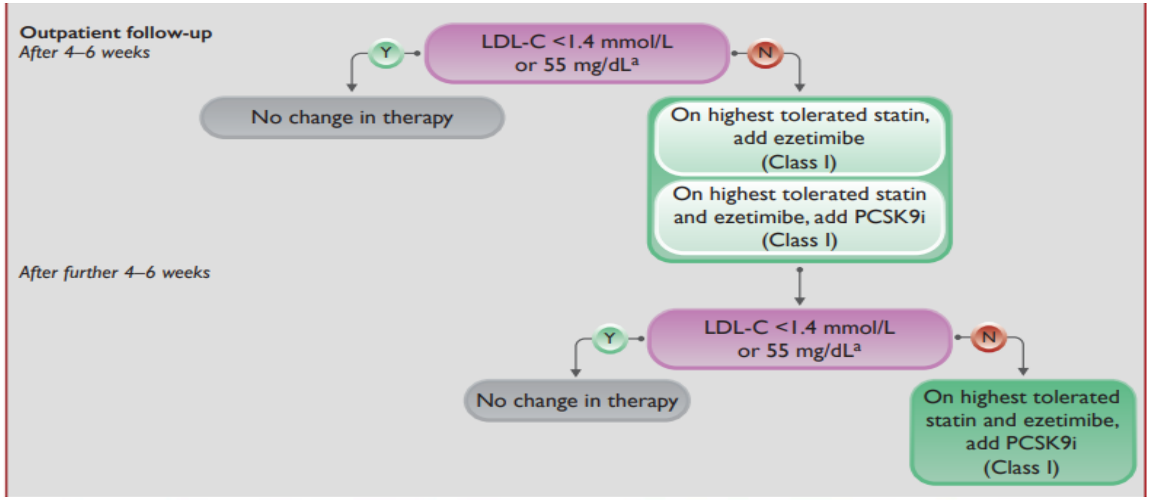
Lipid Lowering Therapies





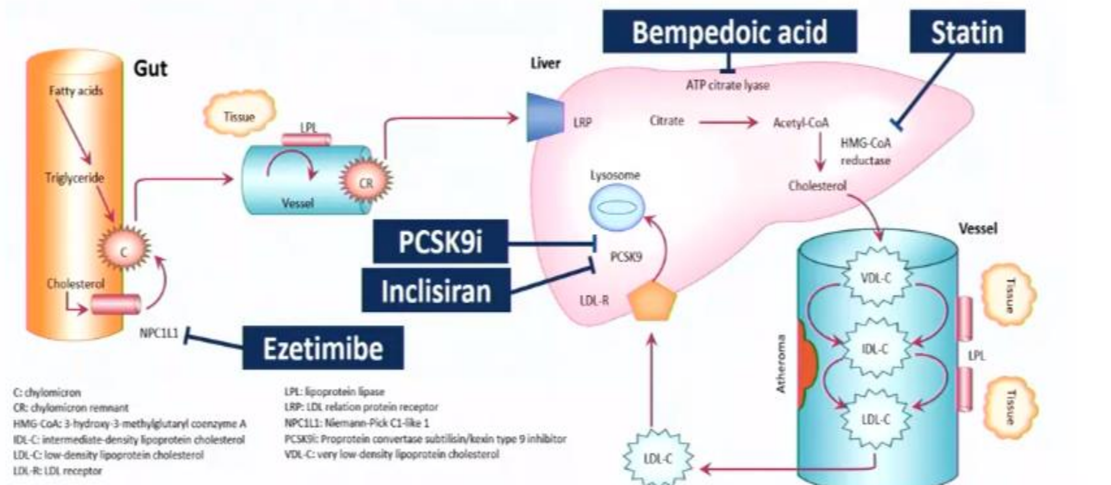


Image adapted from Ryan A, et al. BMJ 2018 and Pinkosky SL, et al. Nat Commun 2016.

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CTM, M/51 (4)

- Successful PCI with stenting to LAD
- Echo showed LVEF 40%
- Discharged with

Aspirin 100 daily
Ticagrelor 90mgBD
Pantoloc 40mg daily
Rosuvastatin 20mg daily
Betaloc 12.5mg daily
Ramipril 1.25mg daily

Explain regarding long-term treatment | Educate about lifestyle modification | Consider mental and emotional health

- Arrange for cardiac rehabilitation – life-style medication, including smoking cessation

Diagnosis

↓

Initial management

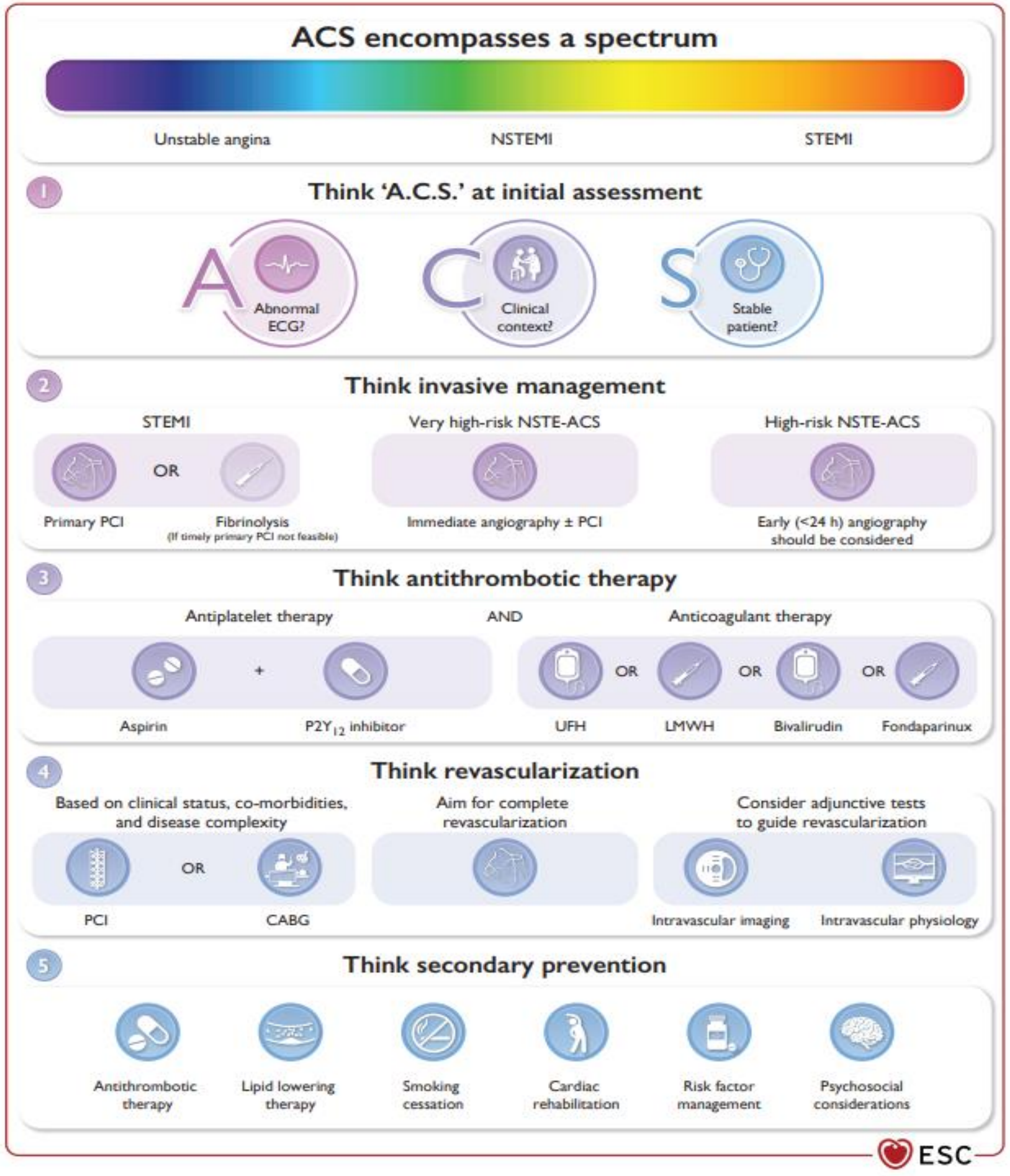
↓

Risk assessment

Invasive vs. Non-invasive

↓

Long-term Rx and 2° prevention



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